Water Retrieval Embedded Attention Network With Multiscale Receptive Fields for Hyperspectral Image Refined Classification

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***Abstract*— In hyperspectral image classification, deep learning (DL) based on abundant training samples has demonstrated its significance in classification performance. However, due to the limitation of available samples and the imbalance/similarity of classes in small-sized datasets, data-driven DL algorithms can hardly extract representative and effective features for interclass classification, and the subtle diagnostic spectral features for intraclass classification are easily covered or lost in the iterative feature extraction (FE). The restricted FE of interclass/intraclass results in the accuracy reduction and performance limitation of refined classification. To mitigate these issues, an attention network with multiscale receptive fields (MRFs) is proposed, embedding an inversion subnet for relative water content retrieval (RWCR). In classification, the three critical parts in the proposed network, namely, MRFs, embedded subnet, and multiple-attention mechanism, are responsible for multiscale feature merging, relative water content (RWC) feature enhancement, and paying attention to bands, channels, and multiscale features, respectively. The ablation studies on small-sized datasets show the accuracy improvements of interclass and intraclass in refined classification, which verifies the effectiveness of critical parts for extracting representative features and taking RWC features as the diagnostic biochemical signature from unbalanced and similar classes. The comparison results with typical DL models demonstrate the superiority of the proposed network. Moreover, the competitive advantage of the proposed network is demonstrated in comparison with traditional and state-of-the-art HSI classification methods.**

***Index Terms*— Hyperspectral image (HSI) classification, multiply attention, multiscale receptive fields (MRFs), relative water content retrieval (RWCR).**

I. Introduction

HYPERSPECTRAL image (HSI) is widely used for the area analysis of interest in Earth observation [1], and the classification of HSI is one of the most important applications in practical research [2], [3]. More and more HSI data have been provided for research with the upgrading of remote sensors. Hyperspectral data are composed of one spectral dimension and two spatial dimensions [4], [5], which provide a large number of signatures of objects for classification. However, hundreds of contiguous narrow bands contain not only abundant features but also redundancy and noise [6]. Moreover, the great majority of HSIs are unlabeled because of the expensive, difficult, and time-consuming labeling process. Therefore, the labeled data are limited in practical application, and only small-sized samples are available for supervised classification [7]–[9].

Small-sized data usually include unbalanced classes and similar classes. Due to the complexity of the Earth's surface, diverse ground objects are collected in HSI data with different contents, which causes the imbalance of class in HSI data [10]. Besides, similar or identical kinds of ground objects usually appear in the same area, and a lot of similar classes are collected in HSI data, such as different subspecies of the same species. Thus, the small-sized data require refined classification algorithms to label the objects. Refined classification requires more accurate classification results than general classification. General classification methods usually focus on interclass distance and have the capacity of classifying different ground objects, which tends to misclassify the similar kinds of ground objects or subspecies of the same species. Refined classification not only improves interclass performance as general classification methods [11] but also improves intraclass performance for similar classes. Although the application of deep learning (DL) in HSI classification is growing rapidly and achieving promising improvements [12], [13], there are still two issues to be addressed in the refined classification on small-sized samples.

First, DL algorithms can hardly extract representative and effective interclass features from unbalanced small-sized HSI datasets due to the number limitation of training samples. Though unsupervised classification methods, such as the K-nearest neighbor [14], the Markov random field [15], the feature learning-based method [16], and the deep belief network (DBN) [17] proposed based on DL or not, achieved satisfactory results without labeled training samples [18], their classification performance is still not competitive with that of supervised methods. The supervised DL methods [19] outperform others, which is greatly attributed to numerous available large-sized labeled data. The convolutional neural network (CNN) [20] was a typical supervised DL algorithm applied to

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HSI that achieved significant improvements in classification. The spectral-based CNN (1-D CNN) [21] extracted features from spectral bands only and discarded spatial information. In order to utilize spatial features in CNNs, 2-D CNN [22] and 3-D CNN [23] methods were introduced into the hyperspectral community and obtained further improvements. The varieties of CNN, such as AlexNet [24], GoogleNet [25], VGGNet [26], DenseNet [27], and ResNet [28], were applied to HSI classification and achieved significant performance. Besides, some deep models based on graph calculation [29], [30], capsule routing [31], and geometry perception [32] are also used in recent works. However, there is still a bottleneck in obtaining satisfactory performance when the small-sized data are classified by CNN models. The bottleneck is essentially caused by the structures of models and the small-sized HSI data that should be processed. In terms of the structure, insufficient utilization of receptive fields (RFs) and attention assignment increases the difficulty of effective interclass feature extraction (FE) [33]. CNN usually takes the HSI data patch as the raw input of the multiple layers, which extracts specific and abstract features from low- and high-level layers, respectively [34]. The size of RFs in the multilayer structure determines the scopes of convolution kernels in FE. CNN is allowed to incorporate contextual information in both spectral and spatial dimensions by RFs, and all RFs of layers share the same size and extract the same-level features [35]. When the input data are small-sized samples, the features in data are limited, and single-scale RFs of the same size are unable to extract various-level features. The scope limitation of single-scale RF weakens the feature representation and decreases the adaptability of multiple layers. Besides, the loss of attention in DL models causes further accuracy reduction. Therefore, many methods of FE or deep network based on attention have been proposed in recent works. It has been demonstrated that the attention mechanism is conducive to selecting effective bands [36] and achieving better results of spectral–spatial classification [37]. Due to the superiority of the attention mechanism, it also has been applied in [38] and [39] and obtains significant improvements. In attention-related works, channel attention proposed in SENet [40], [41] was demonstrated to be effective in improving classification performance. The band attention of HSI data represents the contribution of the original band, which is as important as channel attention. The neglect of attention in spectral–spatial FE [42] not only increases the noise but also weakens the effectiveness of extracted features. In terms of the small-sized data, the unbalanced classes increase the difficulty of interclass FE. The CNN is easily overfitting before extracting effective features from limited training samples, and the imbalance of class limits the available samples furthermore. As a data-driven model, CNN tends to extract features from data-intensive classes, which means that the characteristics of small-number classes are easily covered by that of large-number classes. The unbalanced and small-sized classes require to be classified with limited spatial and similar spectral information, which needs more accurate and representative interclass features than that extracted by single-scale RFs or single attention mechanisms.

Second, subtle diagnostic biophysical–biochemical features are easily covered or lost in the iterative FE, which results in the intraclass accuracy loss on small-sized datasets with similar classes. Various biophysical statuses and different contents of biochemical elements contained in objects generate specific spectral responses [43]. Because of the high spectral resolution of HSI, the subtle differences can be captured by sensors and reflexed in the spectral curve [44]. These biophysical–biochemical features are subtle but diagnostic and critical for refined classification, especially for the intraclass classification of similar classes [45]. For instance, as one of the most common objects in HSI data, vegetation presents diverse spectral curves. Even the same kinds of plants have various spectral curves because of the different structures or element contents (such as water, chlorophyll, and anthocyanin) of leaves. The difference of element content presents sublet change in the spectrum, which is closely related to the growth environment, growth cycle, or growth status of plants [46]. That is, the subtle spectral biophysical–biochemical feature has a strong diagnostic value for similar vegetation in different growth states, which makes it the critical feature of refined intraclass classification. However, subtle biophysical–biochemical features are always covered or lost in multilayer iteration [47], and the characteristic of FE in CNNs is the main reason for the loss of biophysical–biochemical features. The overemphasized mathematical process in multiple layers tends to minimize the calculation loss to maximize the profit of the network, which focuses on enlarging the weights of statistical features. The influences of subtle biophysical–biochemical features on the classification results are reduced after each training epoch [48], [49]. Though CNN models achieve outstanding classification accuracy based on the representative statistical features, there are still some physically/chemically distinguishable classes misclassified. In addition to the multilayer iteration, the downsampling operation in the CNN model (such as maxpooling) also inhibits the expression of subtle diagnostic features. Downsampling preserves the salience features only and discards the subtle features of spectral–spatial dimensions when reducing the resolution of feature maps. Thus, although the subtle biophysical–biochemical features exist in the spectral curves, they are still hardly extracted for classification. The key to overcoming the loss of subtle features is to enhance the extraction of diagnostic biophysical–biochemical features in the iterative process. Although the end-to-end structure of CNN is notable for excluding the human subjectivity and randomness of handcrafted features and extracting data-based features automatically, it is a challenging task to extract specific biophysical–biochemical features and enhance the expression. Therefore, embedding diagnostic biophysical–biochemical features (such as chlorophyll fluorescence and water content) in CNN [50] is remarkable for the reduction of misclassified similar classes and the improvement of intraclass accuracy in refined classification.

To tackle the aforementioned two issues, a water retrieval embedded attention network with multiscale RFs (MRFs) is proposed for refined classification in this article. There are three critical modules responsible for mitigating those

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two issues. First, the MRFs and multiple-attention mechanism are applied for representative and effective interclass FE. Inspired by the multiscale strategy [51], the module of MRFs is taken as the basis of FE in the proposed network. Different from multiscale convolution, the MRFs are feature-oriented rather than data-oriented. The different size of RF means that the overall statistical features and local detail features are extracted on different scales [52]. Moreover, each scale RF can be implemented by multiscale convolution kernels, which means that MRFs can extract diverse scales of features from small-sized samples for classification tasks. To control the contributions of different scales of RFs, scale attention in the multiple-attention module is applied to the RFs for assigning autolearned weights. Besides, the multiple-attention module still contains band attention and channel attention, which are responsible for soft band selection and channel weight control, respectively. Band attention acts on the raw data and generates the global data flow in FE to select informative bands and reduce noise. Channel attention directly affects feature maps by assigning weights on different informative features. Therefore, the modules of MRFs and multiple attentions have the capacity to extract representative and effective interclass features from small-sized data. Second, the relative water content (RWC) is taken as the biochemical diagnostic feature for classification, and an inversion module of RWC retrieval (RWCR) is applied for subtle diagnostic spectral FE. The inversion module is embedded in the proposed network as a subnet, which is trained on totally different HSI datasets. The subnet is responsible for extracting the RWC from HSI data and inserting the RWC features into classification feature maps. In this way, the subtle diagnostic RWC feature is enhanced and taken as biochemical features for intraclass classification. Moreover, the pooling operation is not adopted in the spectral dimension to keep subtle features of the spectrum in the proposed network. The feasibility of critical modules and the effectiveness of the proposed network are demonstrated in experiments.

In this regard, three contributions of this article are summarized as follows.

1) MRFs are proposed and applied to interclass FE as the core part of the proposed network. MRFs can enhance the ability of convolution kernels for FE and fusion from limited training samples, which improves the representation of extracted features. It is demonstrated that the MRFs are significant for accurate FE.
2) A multiple-attention mechanism is introduced for interclass FE. The multiple-attention mechanism pays attention to the informative features extracted from small-sized samples. The scale attention, band attention, and channel attention in the multiple-attention mechanism control the contributions of MRFs, spectral bands, and feature maps of each channel, respectively. It has been demonstrated that the multiple-attention mechanism is supportive of interclass accuracy improvement.
3) An inversion subnet is embedded in the proposed network for the expression enhancement of subtle RWC features in intraclass FE. The introduced diagnostic feature can significantly enhance the expression of RWC features and increases the influence on classification results. The intraclass performance of similar classes with obvious discrimination of water content is improved, and the effectiveness of the subnet is demonstrated.

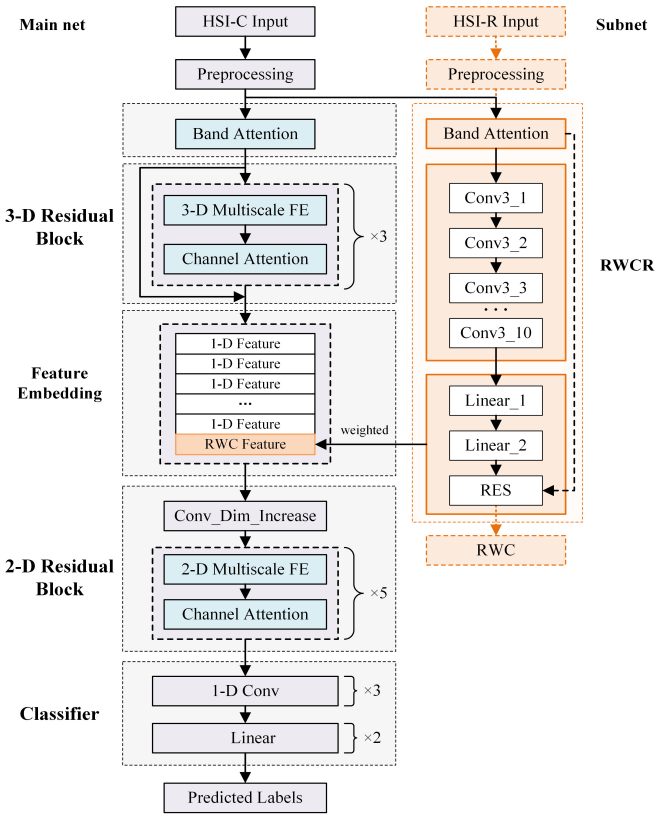

Fig. 1. Flowchart of the proposed network.

The remainder of this article is structured as follows. The proposed network and detailed module implementations are present in Section II. The experimental design, description of datasets, network parameters, and other experimental settings are explained in Section III. The experimental results and discussion are elaborated in Section IV. The conclusion is summarized in Section V.

II. Proposed Method

The proposed network is composed of the main net and the subnet, and the main net contains two critical modules, namely, MRFs and multiple-attention modules. The multiple-attention module is composed of band attention, scale attention, and channel attention. The proposed network is trained on two different types of HSI data for diverse FE. There are four parts in this section to elaborate on the architecture and critical modules of the proposed network.

A. Overview of the Proposed Network

1) Architecture of Proposed Network: The flowchart of two networks in the proposed network is shown in Fig. 1. The left part is the main network containing MRFs and a multiple-attention module, which is responsible for HSI interclass classification. The right part is the inversion subnet, which

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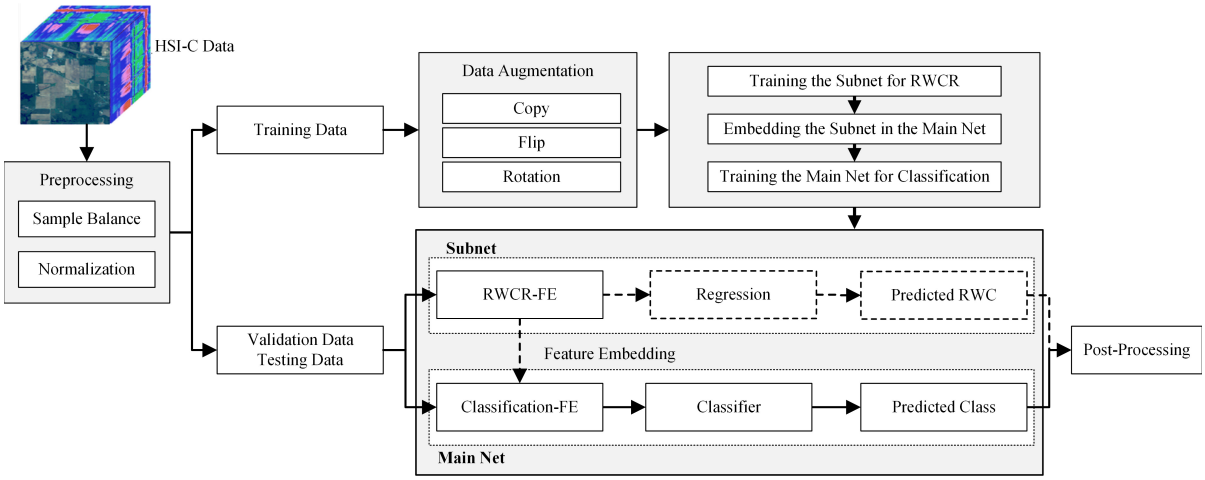

Fig. 2. Training and testing flowchart of the main net.

is responsible for RWCR and inserting the RWC features into the main network for intraclass classification.

As shown in Fig. 1, the main net contains six components for interclass FE between the input and output layers, and the dimensions of feature maps vary with the data flow of components [53]. The two critical modules, MRFs and multiple attentions, are integrated into these components. The main net takes the raw HSI $\mathbf{C} \in \mathbb{R}^{H \times W \times N}$ as the input, where H, W, and N represent the height, width, and band number of HSI, respectively. The first preprocessing component is responsible for data balance, normalization, and augmentation, which can significantly increase the training numbers of small-sized classes and satisfy the requirement of DL training. The second component is the band attention module, which is an important part of multiple-attention modules. Band attention is a soft band selection method [54] for informative FE, which increases or decreases the influence of each band by weighting them in iteration. The next three components compose the residual FE based on the dimensionality-varied mechanism. The third component is the 3-D residual block (3-D RB) that is concatenated by three residual FE units. Each unit consists of a 3-D multiscale FE module and a channel attention module, which are both the implementations of MRFs and multiple-attention mechanisms on 3-D features, respectively. The main task of the 3-D RB is extracting spectral–spatial features from the 3-D HSI cube data and generating 1-D vectors. The input of this block is added to the extracted features to keep the subtle variations of spectra and mitigate information loss in deep layers. It should be mentioned that the residual additions are both in the 3-D RB and 3-D multiscale FE module. The fourth component is a feature embedding block, which is responsible for assembling 1-D vectors generated from 3-D RB and RWCR features into a 2-D feature map. The 1-D vectors are combined in the spatial axis and keeping the spectral direction. Similar to the 3-D RB, the fifth component 2-D residual block is concatenated by five 2-D residual units after a dimension-increase module. Each 2-D residual unit is also composed of a 2-D multiscale FE and a channel attention module. Other operations in 2-D residual block are basically the same as those in 3-D RB, but all are mapped to 2-D features. The last component is a classifier composed of three 1-D convolution layers and two linear layers. It takes the output of the 2-D residual block as the input and outputs the classification results.

The subnet for enhancing the expression of RWC features and improving the intraclass accuracy is described in Section II-D.

2) Training and Testing of the Proposed Network: There are two steps for the training of the proposed network, and two different types of HSI data are used in the training process. The first-type data (HSI-C in Fig. 1) is a general HSI for classification, and the second-type data (HSI-R in Fig. 1) is an HSI with RWC labels for inversion subnet training. The HSI-R dataset is similar to the HSI-C dataset, both of which are hyperspectral datasets. The difference between them is that the labels of HSI-R are not the classes of ground truth as HSI-C but the RWC values of the corresponding spectra. Due to the similarity of HSI-R and HSI-C data structure, the subnet and the main net are trained according to the training of general supervised networks. The training steps are given as follows.

The first is training the inversion subnet on HSI-R data. During the supervised training, the spectral–spatial data and RWC labels are taken as the input and the output of the subnet, respectively. Based on the supervised information of RWC, the subnet can extract the RWC features for regression. The trained subnet has the capacity to retrieve RWC from HSI data.

The second is embedding the subnet in the main net and starting training the main net for classification. The parameters of the trained subnet are not updated during the training of the main net. Based on the embedded RWC features and supervised information of classes, the main net can be trained to extract more accurate features for classification.

The training and testing processes of the main net are shown in Fig. 2. Before the training of the main net, the subnet for RWCR is trained on HSI-R data, and the well-trained subnet is embedded in the main net. Then, the main net is trained on the HSI-C data for classification. That is, the subnet of RWCR is a functional plug-in module for retrieving RWC in the main net. As for the training of the main net, the small-sized HSI-C data are preprocessed for class balance and data normalization.

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The numbers of training data are enlarged by copying, flipping, and rotating to satisfy the DL training requirement. All augmented data are taken as the input of the proposed network. During the training process, the parameters of the main net are updated by HSI-C data, and the parameters of the subnet remain the same for RWC FE. As for the validation and testing, all corresponding data are input into the trained network for classification. In the subnet, the RWCR-FE (in Fig. 2) works, but other parts of the subnet are not activated. The features extracted by the subnet and the main net are merged for classification. Specifically, the RWC features extracted by the subnet are merged with the features extracted by the main net and input into the classifier for prediction and postprocessing.

The loss function $E(\omega)$ of the main net is set as the cross-entropy loss, where ω is the parameter of the network. Unless otherwise specified, all the same symbols represent the same meaning in the following of this article. The loss function can be formulated as

$$E(\omega) = -\sum_{i=1}^{n} y_i \ln(\hat{y}_i) \tag{1}$$

where n is the number of classes in the ground truth, and y_i and $\hat{y}_i$ are the ith truth label and the corresponding element in the softmax output, respectively. $\hat{y}_i$ can be calculated as

$$\hat{y}_i = \frac{e^{a_i}}{\sum_{j=1}^{n} e^{a_j}} \tag{2}$$

where a_i and a_j are the ith and jth values in the input of the classifier. The parameters are updated after each epoch, which can be formulated as

$$\omega_{i+1} = \omega_i - \eta \nabla \omega = \omega_i - \eta \frac{\partial E(\omega)}{\partial \omega} \tag{3}$$

where η is the dynamic learning rate. η varies with the training epochs s and the overall accuracy (OA) of validation, which is conducive to obtaining a relative equilibrium between the converge and loss of DL model. η is reduced when the validation OA has not been increased for two training epochs. η remains unchanged when it is reduced to a predefined minimum value in iteration to ensure the convergence of the proposed network. The dynamic learning rate can be formulated as

$$\eta_{s+1} = \eta_s \times f_\eta \tag{4}$$

where f_η is the factor of reduction.

B. Multiscale Receptive Fields in Main Net

The detailed MRFs are shown in Fig. 3. An MRF consists of two parts for FE. The left part $f_D(\cdot)$ is a diagnostic block (DG) of spectral FE, which consists of convolution kernels (Conv1) with size 1, a batch normalization layer $f_{\text{Norm}}(\cdot)$, a rectified linear unit (ReLU) layer $f_{\text{ReLU}}(\cdot)$, and a spatial pooling layer $f_{\text{Pool}}(\cdot)$. The left part can be formulated as

$$f_D = f_{\text{Pool}}(f_{\text{ReLU}}(f_{\text{Norm}}(\text{conv}(\mathbf{x}_l, \mathbf{k}_{l+1})))) \tag{5}$$

where $\mathbf{x}_l$ is the lth layer feature map and $\mathbf{k}_{l+1}$ is the kernels of Conv1 for extracting diagnostic spectral features and retaining the dimensions of input feature maps. The activation function

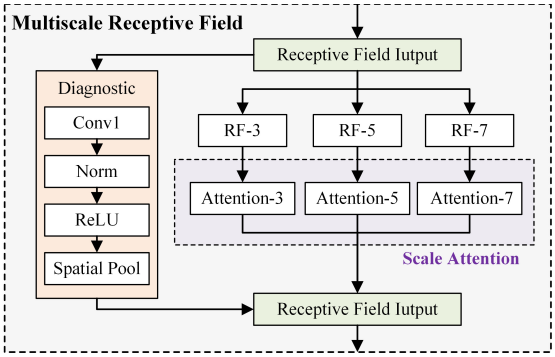

Fig. 3. Structure of MRFs.

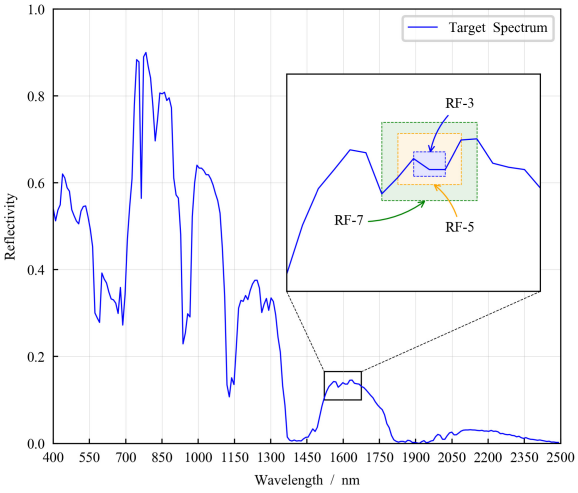

Fig. 4. Structure of MRFs on spectral dimensions.

ReLU is unsaturated and faster than other saturation methods. The right part $f_{\text{RF}s}(\cdot)$ is an FE block based on MRFs, namely, RF-3, RF-5, and RF-7. The RFs are set as 3, 5, and 7 on both spatial and spectral dimensions, and each of them is weighted by corresponding scale attention $f_{\text{AS}}(\cdot)$. To reduce the parameters of RFs, all RFs are implemented by convolution kernels with three parameters on each dimension. The right part can be formulated as

$$f_{\text{RF}s} = \text{sum}(f_{\text{AS}}(f_{\text{RF3}}(\mathbf{x}_l), f_{\text{RF5}}(\mathbf{x}_l), f_{\text{RF7}}(\mathbf{x}_l))). \tag{6}$$

The structures of MFRs on spectral and spatial dimensions are shown in Figs. 4 and 5, respectively. Different-scale RFs extract various features, and each of them contains a different characteristic barycenter. The diversity of feature maps makes it possible to express and map selectively in the DL model. More importantly, it keeps the subtle changes of features in different RFs, which improves the sensitivity of the DL model to interclass characteristics.

The detailed RF-7 is shown in Fig. 6. Inspired by ResNet [28], there are three bottleneck-shape FE units in RF-7, namely, two dilated [55] and one stacked convolution units. Each of them contains a residual operation to retain the input information. The Conv1 operations in the start and end of the bottleneck are responsible for dimension reduction

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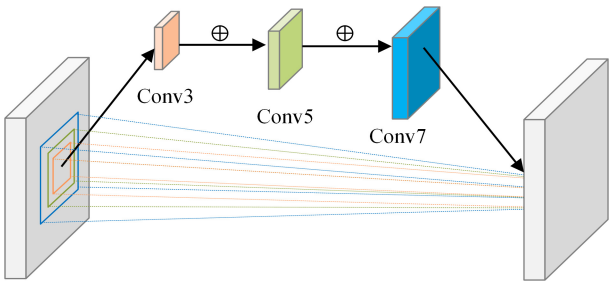

Fig. 5. Structure of MRFs on spatial dimensions.

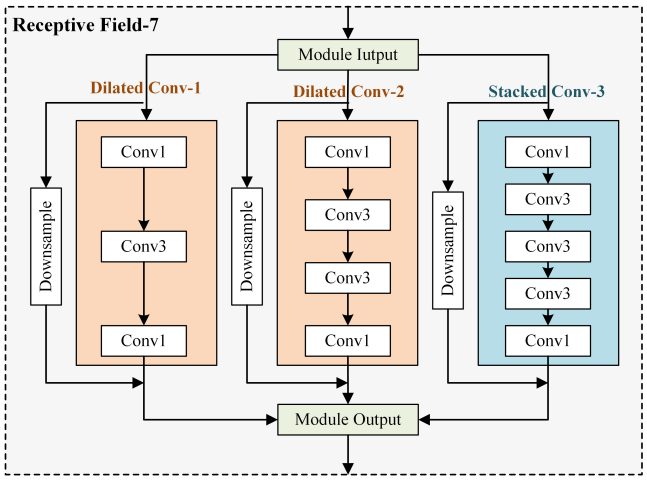

Fig. 6. Data flow of RF-7.

and increase, which reduce numerous training parameters as in ResNet. Since the MRFs have been applied to both 3-D and 2-D residual blocks of the proposed network, the dilated and stacked convolution operations are both used for FE. For 3-D RBs, the $7 \times 7 \times 7$ convolution can be decomposed into three stacked $3 \times 3 \times 3$ regular convolutions (Conv3) with nonlinearity injected. Similarly, the $7 \times 7 \times 7$ convolution also can be decomposed into two or one concatenated $3 \times 3 \times 3$ dilated convolution, with the spacing between the kernel points (namely, the dilation value) of 2 or 3, respectively. The dilated convolution allows for the exponential increase in the field of view without the decrease in dimensions. The dilated and regular 3-D convolutions both can be formulated as

$$\mathbf{x}_{l+1}^{i,j,k} = \sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{n=1}^{N} \mathbf{x}_{l}^{g(h),g(w),g(n)} \cdot \mathbf{k}_{l+1}^{h,w,n} \tag{7}$$

$$g(h) = i + h \cdot (d-1) \tag{8}$$

where H, W, and N indicate the height, width, and length of the $(l+1)$th layer convolutional kernel. $\mathbf{k}_{l+1}^{h,w,n}$ represents the element of the kernel, with (h, w, n) being the coordinate of element position. $\mathbf{x}_{l}^{g(h),g(w),g(n)}$ is the corresponding input element in the feature map, with $(g(h), g(w), g(n))$ being the coordinate. $\mathbf{x}_{l+1}^{i,j,k}$ is the corresponding output of the convolution, with (i, j, k) being the coordinate. Moreover, $d(d \geq 1)$ in (8) is the dilation value, which is set as 1 in the regular convolution and other corresponding values in the dilated convolution, respectively. The 2-D residual blocks follow the same criteria as 3-D RBs, and the convolution is given by

$$\mathbf{x}_{l+1}^{i,j,k} = \sum_{h=1}^{H}\sum_{w=1}^{W} \mathbf{x}_{l}^{g(h),g(w)} \cdot \mathbf{k}_{l+1}^{h,w}. \tag{9}$$

As the sum of those outputs of data flows, the output of RF-7 is formulated as

$$f_{\mathrm{RF7}} = \sum_{i=1}^{3} (\mathrm{ds}_i(\mathbf{x}_l) + \mathrm{conv}_i(\mathbf{x}_l)) \tag{10}$$

where $\mathrm{conv}_i(\cdot)$ is the ith convolution operation, namely, Dilated Conv-1 (DC-1), Dilated Conv-2 (DC-2), and Stacked Conv-3 (SC-3), and $\mathrm{ds}_i(\cdot)$ is the corresponding downsample operation for feature mapping of input and output. RF-5 and RF-3 have a similar module structure to RF-7 but with one and zero dilate convolution units, respectively. Therefore, MRFs in the proposed network extract the statistical and accurate interclass features by focusing on global and local characteristics and reduce the computational costs without spectral pooling. It mitigates the interclass feature loss in iterative training and improves the efficiency of the model.

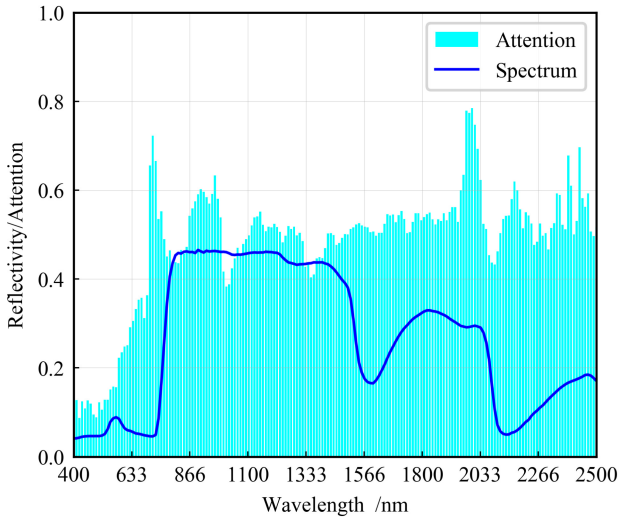

Fig. 7. Attention for spectral information.

C. Multiple-Attention Mechanism in Main Net

As described above, there are three-type attention modules applied to the proposed network, namely, band attention at the beginning of the network, scale attention in the MRFs, and channel attention after each MRF. The attention mechanism is inspired by human visual attention [56] and assigns different weights to the feature maps. The importance and contributions of HSI bands, multiscale features, and feature map of each channel are different for classification. The band attention is a soft band selection method, which selects the most informative and discriminative bands by weighting instead of retaining or discarding. Band weight controls contributions to emphasize the information and suppress noises [57]. The attention on

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the spectrum is shown in Fig. 7. The band attention can be formulated as

$$\mathbf{B}_A = f_{AB}(\mathbf{C}) = \mathbf{A}^{(\text{tiled})} \cdot \mathbf{C} \tag{11}$$

where $\mathbf{C}$ and f_{AS} is the HSI data and band attention function and $\mathbf{B}_A$ is the corresponding output. The weights of band attention are initiated randomly. The length of attention $\mathbf{A}$ is equal to the band number, and the values of $\mathbf{A}$ are usually initialized in the range of [0, 1] in practice. Besides, $\mathbf{A}$ is an all-band-attention vector for each weight of the band, which should be tiled as $\mathbf{A}^{(\text{tiled})}$ for all spatial pixels and mapping the feature dimensions. Similar to band attention, scale attention is for weighting the various features of MRFs to select the representative ones. All values of scale attention are initialized as that of band attention. The channel attention is implemented based on SE-block [41], which is not totally the same as band attention or channel attention. It extracts the global feature from the current feature map $\mathbf{x}_l \in \mathbb{R}^{H\times W\times N}$ by the spatial average pooling, which is formulated as

$$\mathbf{W}_A = f_{\text{S}}\big(f_{\text{ReLU}}\big(f_{\text{ap}}(\mathbf{x}_l)\big)\big) \tag{12}$$

$$f_{\text{ap}}(\mathbf{x}_l) = \frac{1}{H\times W\times N}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{n=1}^{N}\mathbf{x}_l^{h,w,n} \tag{13}$$

where $f_{\text{S}}(\cdot)$ is the sigmoid function and $f_{ap}(\cdot)$ is the average pooling function. Other symbols remain the same meanings as (5) and (7). The channel attention $\mathbf{W}_A$ is then multiplied by $\mathbf{x}_l$ to generate the attention-based feature map.

All trainable attention weights are updated in training and learn to pay attention to spectral, scale, and channel domains so that the significance of features in different domains can be discriminated and the informative features can be selected for improving the interclass classification accuracy.

D. Embedded Subnet

The main task of the subnet is different from that of MRFs and multiple-attention modules in the main net. MRFs and multiple attentions are applied to the main net of the proposed network for extracting representative and effective features, and improving the interclass accuracy of small-sized datasets. The subnet is responsible for the expression enhancement of RWC features in FE to improve the intraclass accuracy of small-sized datasets.

RWC is derived from the vegetation inversion model, which aims at retrieving the element content of the leaf, including the water content. RWC is a relative concept, which only focuses on the RWC of ground objects. Each object has unique absorption and scattering properties, which can be captured by HSI sensors with high spectral resolution. The spectral curve of each ground object reflects the inherent biophysical or biochemical properties [58], including water content in their surface or interior. Due to the abundance of plants in HSI data, it is convenient to compare the plants with other ground objects in Earth observation [59]. Consequently, the concept of RWC is used for subtle FE. In the DL-based networks of HSI classification, the models emphasize the statistical features to minimize the loss and maximize the profit of training. Some subtle diagnostic features are overwhelmed by those statistical features in this mathematical process. However, introducing the RWC into the deep models as the biochemical property is able to enhance the subtle diagnostic features for classification.

Therefore, the subnet is trained for RWCR and embedded in the main net for classification in the proposed model. The significance of the subnet for RWCR in the main net is to extract the RWC features from spectra and enhance the expression in classification. The extracted RWC features in the main net can effectively reduce misclassified plants that are in different growth statues or the misclassified objects that are physically/chemically distinguishable (such as vegetation and stones) [60]. The structure of the subnet is much simpler than that of the main net. Similar to the main net, there are preprocessing and band attention modules before FE by convolution. There are ten stacked 3-D convolution layers for FE and two linear layers for outputting. Each convolution layer contains 3-D kernels, batch normalization, and the ReLU function, which can be formulated as

$$\mathbf{x}_{l+1} = f_{\text{ReLU}}(f_{\text{Norm}}(\text{conv}(\mathbf{x}_l, \mathbf{k}_{l+1}))). \tag{14}$$

It is noteworthy that the spectral index of red slope edge (RES) based on our previous work [54] is also used for RWC feature fusion or calibration in the strategic layer. The RES was proposed based on prior knowledge and spectral characteristics, which can be formulated as

$$\text{RES} = 2\sin\left(\frac{\Delta R_{760,650}}{\Delta\lambda_{760,650}}\right) \tag{15}$$

where $\Delta R_{760,650}$ is the difference of spectral reflectance values of 760 and 650 nm, with $\Delta\lambda_{760,650}$ being the corresponding difference of wavelength. In practice, RES can be simply calculated as

$$\text{RES} = 2\sin(R_{760} - R_{650}). \tag{16}$$

The loss function $E(\omega)$ of the subnet is set as the mean squared error (mse) loss, where ω is the parameter of the network. The loss function can be formulated as

$$E(\omega) = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2 \tag{17}$$

where m is the number of training samples. Other symbols in (14)–(17) remain the same meanings as the above formulations.

The main net is a typical deep classification model that focuses on interclass classification accuracy only. The subnet is only trained by the RWC datasets as described in the aforementioned training steps and embedded in the main net for intraclass FE. During the training of the main net, the subnet focuses on the RWC FE and keeps the parameters not been updated to enhance the expression of subtle diagnostic features.

The diagnostic RWC features generated by the subnet are assembled in the main net for intraclass classification, and other feature maps of the main net are responsible for interclass classification. Therefore, the proposed network significantly improves the performance of intraclass and interclass in refined classification.

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III. Experimental Design and Settings

A. Experimental Design

According to the description of training and testing steps in Section II, the proposed network was trained on two different types of datasets (HSI-R and HSI-C), respectively. Specifically, the subnet is trained on the HSI-R dataset that takes the RWC labels as the ground truths. The main net is trained on the HSI-C dataset that takes the class labels as the ground truths. The subnet was trained and tested on HSI-R first, and then, the well-trained subnet is embedded in the main net for the training on HSI-C. Thus, the experiments were divided into four parts in this article: 1) the performance evaluation of the subnet; 2) the interclass and intraclass performance analysis of the proposed network in an ablation study; 3) the superiority verification of the proposed network in comparison with typical DL algorithms; and 4) the superiority verification of the proposed network in comparison with state-of-the-art HSI classification methods. The experiments were conducted on small-sized datasets for qualitative and quantitative performance evaluations.

B. Hyperspectral Datasets

To fulfill the experimental requirements, HSI-R contains three leaves' datasets for RWCR of the subnet, and HSI-C contains three public datasets for the classification of the main net. The leaves' datasets contain two datasets collected in the laboratory (Leaves-1 and Leaves-2) for qualitative evaluation and one public dataset Lopex for quantitative evaluation. The three public classification datasets are Indian Pines (IP), Salinas Valley (SV), and Pavia University scene (PU). In order to evaluate the performance of refined classification and verify the accuracy improvements of interclass/intraclass features, 24 objects were selected from 41 ground objects of IP, SV, and PU to generate eight new subdatasets for HSI-C. Four criteria were adopted in the generating process, and two subdatasets were generated according to each criterion. The first criterion was the same plants of pure species, which contained Corn and Lettuce datasets. The second criterion was the same plants of mixed species, which contained Grass and Brocoli–Weed (BW) datasets. The third criterion was the mixed plants, which contained Hay-Oats-Woods (HOW) and Hay-Soybean-Woods (HSW) datasets. The fourth criterion was the plants and nonplants, which contained Meadow-Tree-Brick (MTB) and Meadow-Tree-Soil (MTS) datasets.

1) *Leaves-1 Dataset:* Leave-1 was collected in the laboratory for qualitative evaluation of RWCR. The collection equipment was the HeadSpecIII imaging system with two sensors. There are 812 bands captured by two sensors in the images, and the effective band range is 400–800 nm. The spectral resolution is smaller than 7 nm. The images of Leaves-1 are shown in Fig. 8. All leaves in Fig. 8 were collected from the identical plant and arranged from left to right according to RWC. Fig. 8(a) shows the fresh leaf with the most abundant water content, which has the highest-level RWC. The main part of Fig. 8(b) has the second-level RWC, which has less water content than Fig. 8(a). The third-level RWC is in Fig. 8(c), which is drier than the second-level leaf. The fourth-level RWC is in the right bottom of Fig. 8(b), which is in abnormal growth status. The lowest-level RWC is in Fig. 8(d), which is a simulated leaf made of plastic. There are five average spectral data selected from the above area for the qualitative evaluation.

Fig. 8. Leaves-1 data images. (a) Leaf sample with the highest-level RWC. (b) Leaf sample with the second-level and fourth-level RWC. (c) Leaf sample with the third-level RWC. (d) Leaf sample with the lowest-level RWC.

Fig. 9. Leaves-2 data images.

2) *Leaves-2 Dataset:* Leave-2 data were collected on the campus of the Harbin Institute of Technology for the qualitative evaluation of RWCR. The Leaves-2 image is shown in Fig. 9. There was a round dry area of the lawn because of the concentration of hot water pipes. Leaves-2 was collected by the SPECIM VNIR sensor with the spectral range being 400–1000 nm. There are 490 $\times$ 320 pixels in the data, and each pixel has corresponding 204 spectral bands, with the spectral resolution being 3 nm.
3) *Lopex Dataset*: Lopex is a public leaf property dataset collected in Ispra, Italy, which is used for the quantitative evaluation of RWCR. There are more than 20 properties in this dataset, and the leaf equivalent water thickness (EWT) was selected as the RWC label. Lopex contains 66 classes of leaves, including the same or different species. Each of them contains five labeled samples. All $66 \times 5 = 330$ samples have 2101 bands in the range of 400–2500 nm, and the spectral resolution is 1 nm. The bands of Leaves-1, Leaves-2, and Lopex in the range of 400–800 nm were selected and downsampled to 200 for mutual matching. The downsampled spectral curves of classes in Lopex are shown in Fig. 10.
4) *IP Dataset*: The IP dataset is the HSI of IP agricultural fields in Northwestern Indiana, USA, which was gathered by the AVIRIS sensor in 1992. The HSI data contains 145 $\times$ 145 pixels with the spatial resolution being 20 m. AVIRIS gathered 220 bands in the range of 400–2500 nm, but 20 of them in the IP dataset were

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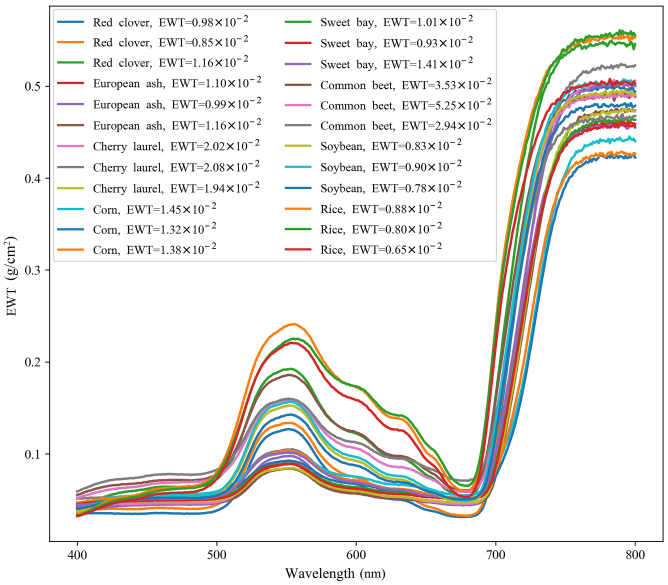

Fig. 10. Lopex spectral curves.

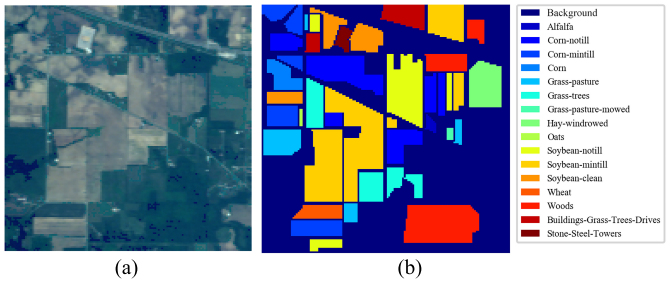

Fig. 11. IP dataset: (a) false-color image and (b) ground-truth image.

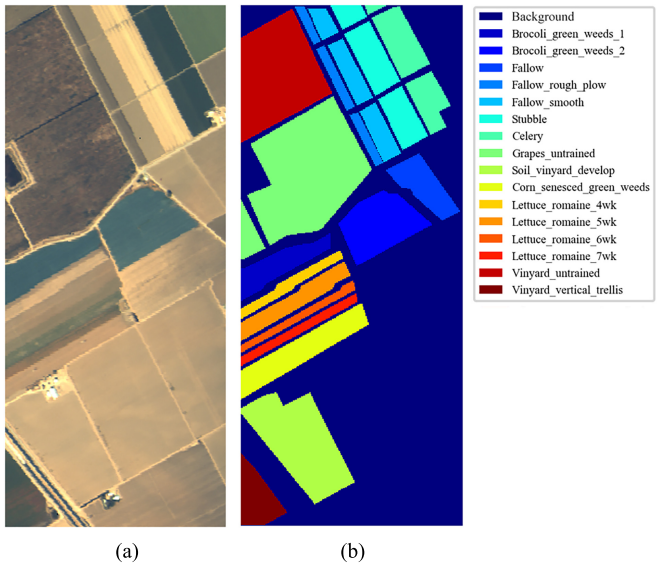

Fig. 12. SV dataset: (a) false-color image and (b) ground-truth image.

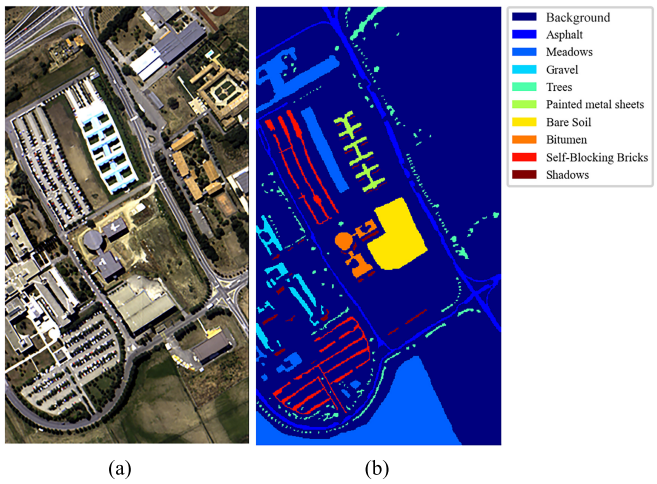

Fig. 13. PU dataset: (a) false-color image and (b) ground-truth image.

removed due to the invalidation or noise. The IP dataset contains 16 mutually exclusive ground objects, as shown in Fig. 11.

5) *SV Dataset*: The SV dataset is the HSI of the SV in California, USA, which was also gathered by the AVIRIS sensor. SV data contain 512 × 217 pixels with a high spatial resolution being 3.7 m. As with the IP dataset, 20 water absorption bands in the range of 400–2500 nm were removed, and the left 200 bands were available. The SV dataset contains 16 mutually exclusive ground objects, as shown in Fig. 12.
6) *PU Dataset*: The PU dataset is the HSI of the campus of PU in Northern Italy, which was acquired by the ROSIS sensor in 2001. The HSI data contain 610 × 340 pixels with the spatial resolution being 1.3 m. The PU dataset contains 103 spectral bands in the range of 430–860 nm after removing some noisy bands. There are nine mutually exclusive ground objects in the PU dataset, as shown in Fig. 13.
7) *Eight Subdatasets:* 24 classes were selected from 41 ground objects in the three public datasets (IP, SV, and PU) and divided into eight subdatasets according to four criteria. The 24 classes contained plants, mixed plants, soil, and other artificial objects. Various classes were combined as new efficient datasets for the performance evaluation of the proposed network. Specifically, in the first criterion of the same plants of pure species, the Corn dataset contains corn-notill, corn-mintill, and corn classes, and the Lettuce dataset contains lettuce classes from four to seven weeks. In the second criterion of the same plants of mixed species, the Grass dataset contains three mixed-specie classes, namely, grass-pasture, grass-trees, and grass-pasture-mowed. The BW dataset contains two mixed-species classes of brocoli_weeds_1 and brocoli_weeds_2. In the third criterion of mixed plants, the HOW dataset contains hay-windrowed, oats, and woods plants, and the HSW dataset contains hay-windrowed, soybean-notill, and wood plants. In the fourth criterion of plants and nonplants, the MTB dataset contains meadows, trees, and self-blocking bricks, and the MTS dataset contains meadows, trees, and bare soil. The combinations of

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TABLE I
TRAINING NUMBERS OF GROUND-TRUTH CLASSES IN THE IP DATASET

Class	Total	Training		T	Augmentation	
		15%	10%		15%	10%
Alfalfa	46	7	5	30	1470	1050
Corn-notill	1428	215	143	1	1505	1001
Corn-mintill	830	125	83	2	1750	1162
Corn	237	36	24	6	1512	1008
Grass-pasture	483	73	49	3	1533	1029
Grass-trees	730	110	73	2	1540	1022
Grass-pasture-mowed	28	5	3	50	1750	1050
Hay-windrowed	478	72	48	3	1512	1008
Oats	20	3	2	75	1575	1050
Soybean-notill	972	146	98	2	2044	1372
Soybean-mintill	2455	369	246	1	2583	1722
Soybean-clean	593	89	60	3	1869	1260
Wheat	205	31	21	7	1519	1029
Woods	1265	190	127	1	1330	889
Buildings-Grass-Trees-Drives	386	58	39	4	1624	1092
Stone-Steel-Towers	93	14	10	15	1470	1050
Total Samples	10249	1543	1031		26586	17794

different spices or objects are used for the effectiveness of RWCR and the interclass/intraclass performance evaluation.

8) *Houston Datasets*: The Houston 2013 dataset [61] is the HSI of the University of Houston, which was used for the scientific purpose of the 2013 IEEE GRSS Data fusion Contest. It was collected by the ITRES-CASI 1500 sensor with 144 bands. The spatial resolution is 2.5 m with 349 $\times$ 1905 pixels. The spectral resolution is in the range of 380–1050 nm. There are 15 classes in the dataset.

Due to the effectiveness of data augmentation for DL methods, all of the aforementioned datasets are augmented after normalization to enlarge the datasets. First, the small-number classes of three public datasets (IP, SV, and PU) were copied T times to balance the samples. Second, all balanced classes were copied once and flipped horizontally and vertically along the spectral axis, respectively. Moreover, the data were also rotated by 90°, 180°, and 270° along the spectral axis. That is, the numbers of training samples were enlarged to seven times after data enhancement. Accordingly, as for RWCR in the subnet, there are 1848 samples for training, 33 samples for validation, two image datasets for qualitative testing, and 33 samples for quantitative testing. As for the classification of the proposed network, the aforementioned datasets and corresponding numbers of training samples are shown in Tables I–IV. There are 26 586, 119 245, and 69 895 labeled samples of IP, SV, and PU for 15% training, respectively, and 17 794, 19 047, and 9345 labeled samples of them for 10% of IP, 2% of SV, and 2% of PU training. As for the eight subdatasets, there are 3150, 490, 2310, 210, 4690, 350, 420, and 700 labeled samples for training. As for the Houston dataset, we have selected 10%, 20%, and 30% samples from the training set for the training process.

C. Experimental Settings

According to the experimental design, the corresponding settings of the four parts are given as follows.

TABLE II
TRAINING NUMBERS OF GROUND-TRUTH CLASSES IN THE SV DATASET

Class	Total	Training		T	Augmentation	
		15%	2%		15%	2%
Brocoli_green_weeds_1	2009	302	41	2	4228	574
Brocoli_green_weeds_2	3726	559	75	2	7826	1050
Fallow	1976	297	40	3	6237	840
Fallow_rough_plow	1394	210	28	3	4410	588
Fallow_smooth	2678	402	54	2	5628	756
Stubble	3959	594	80	2	8316	1120
Celery	3579	537	72	2	7518	1008
Grapes_untrained	11271	1691	226	2	23674	3164
Soil_vinyard_develop	6203	931	125	2	13034	1750
Corn_green_weeds	3278	492	66	2	6888	924
Lettuce_romaine_4wk	1068	161	22	2	2254	308
Lettuce_romaine_5wk	1927	290	39	2	4060	546
Lettuce_romaine_6wk	916	138	19	2	1932	266
Lettuce_romaine_7wk	1070	161	22	2	2254	308
Vinyard_untrained	7268	1091	146	2	15274	2044
Vinyard_vertical_trellis	1807	272	181	3	5712	3801
Total Samples	54129	8128	1236		119245	19047

TABLE III
TRAINING NUMBERS OF GROUND-TRUTH CLASSES IN THE PU DATASET

Class	Total	Training		T	Augmentation	
		15%	2%		15%	2%
Asphalt	6631	995	133	1	6965	931
Meadows	18649	2798	373	1	19586	2611
Gravel	2099	315	42	3	6615	882
Trees	3064	460	62	2	6440	868
Painted metal sheets	1345	202	27	4	5656	756
Bare Soil	5029	755	101	1	5285	707
Bitumen	1330	200	27	4	5600	756
Self-Blocking Bricks	3682	553	74	2	7742	1036
Shadows	947	143	19	6	6006	798
Total Samples	42776	6421	858		69895	9345

TABLE IV
TRAINING NUMBERS OF GROUND-TRUTH CLASSES IN EIGHT SUBDATASETS

Data set	Class	Total	Training	T	Augmentation
Corn	Corn-notill	1428	50	1	350
	Corn-mintill	830		2	700
	Corn	237		6	2100
Lettuce	Lettuce_4wk	1068	10	2	140
	Lettuce_5wk	1927		1	70
	Lettuce_6wk	916		2	140
	Lettuce_7wk	1070		2	140
Grass	Grass-pasture	483	10	2	140
	Grass-trees	730		1	70
	Grass-pasture-mowed	28		30	2100
BW	Brocoli_weeds_1	2009	10	2	140
	Brocoli_weeds_2	3726		1	70
HOW	Hay-windrowed	478	10	3	210
	Oats	20		63	4410
	Woods	1265		1	70
HSW	Hay-windrowed	478	10	3	210
	Soybean-notill	972		1	70
	Woods	1265		1	70
MTB	Meadows	18649	5	1	35
	Trees	3064		6	210
	Self-Blocking Bricks	3682		5	175
MTS	Meadows	18649	10	1	70
	Trees	3064		6	420
	Bare Soil	5029		3	210

In the first part of the experiments, the subnet was trained and tested for performance evaluation. Three different leaves' datasets (Leaves-1, Leaves-2, and Lopex) were selected for the

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TABLE V
MAJOR CONVOLUTION PARAMETERS OF MRFs IN 3-D RB

DG	RF-3	RF-5		RF-7			
	SC	DC-1	SC-2	DC-1	DC-2	SC-3	
1×3×3	1×1×1	1×1×1	1×1×1	1×1×1	1×1×1	1×1×1	×3
	3×3×3	3×3×3	3×3×3	3×3×3	3×3×3	3×3×3	
	1×1×1	1×1×1	3×3×3	1×1×1	3×3×3	3×3×3	
			1×1×1		1×1×1	3×3×3	
						1×1×1	

TABLE VI
MAJOR CONVOLUTION PARAMETERS OF MRFs IN 2-D RESIDUAL BLOCK

DG	RF-3	RF-5		RF-7			
	SC	DC-1	SC-2	DC-1	DC-2	SC-3	
3×3	1×1	1×1	1×1	1×1	1×1	1×1	×5
	3×3	3×3	3×3	3×3	3×3	3×3	
	1×1	1×1	3×3	1×1	3×3	3×3	
			1×1		1×1	3×3	
						1×1	

training and testing of the subnet. We selected 80% labeled samples of Lopex for training and 10% for validation. Besides, we also selected Leaves-1 and Leaves-2 for qualitative testing and the left 10% of Lopex for quantitative testing.

In the second part, we had tested the effectiveness of each key module and the influence of training sample size and spatial size on classification performance. In the key modules test, the classification performance of the proposed network with MRFs being the core FE part was considered as the benchmark. The proposed network without the multiple-attention mechanism and without the subnet was tested for the performance of interclass and intraclass classifications, respectively. The ablation study was carried on the aforementioned eight subdatasets in four categories, which were generated from the three public classification datasets. The numbers of training samples were in the range of 5–50. The validation sample numbers were 1% for MTB and MTS and 10% for the others. The left labeled samples were for testing.

In the third part, to verify the superiority of the proposed network, we had selected five different typical and popular DL algorithms, namely, AlexNet [24], GoogleNet [25], VGGNet [26], DenseNet [27], and ResNet [28] for comparison. We selected a dataset from each category of the second part as experimental datasets, and all algorithms were trained, validated, and tested on the four datasets with the same ratio as the second part.

In the fourth part, the comparison experiments with state-of-the-art HSI classification methods were divided into four parts to evaluate the superiority of the proposed method according to the size of the training samples and datasets. The first part of classification methods contained SVM, RF, MLP, 2-D CNN, and 3-D CNN in [62] and [63]; 15% of labeled samples of each dataset were selected for training. The second part of classification methods contained SMBN [64], DFFN [65], DHCNet [66], SSRN [67], SSN [68], and SSAN [68]; and 10% of IP, 2% of SV, and 2% of PU were selected for training, 10% of labeled data were for validation, and the left was for testing. The third part of classification methods contained SVM, 3-D CNN, SSRN, RN-FSC, and DCFSL in [69]. Five labeled samples per class in IP, SV, and PU datasets were selected for training. The fourth part of classification methods contained the above SVM, 2-D CNN, 3-D CNN, SSRN, and SSAN for further validation on the Houston dataset. The training and testing data are selected from the training set and the testing set.

The major convolution kernels of the main net in the proposed network are shown in Tables V and VI. Table V shows the kernels of 3-D MRFs in the 3-D RBs, containing DG, RF-3, RF-5, and RF-7, as described in Section II. Table VI shows the corresponding convolution kernels of 2-D residual blocks. Besides, some other hyperparameters are set empirically as follows. All weights of the proposed network were initialized randomly and optimized by the Adam optimizer. The dynamic learning rate was initialized to 0.01 to control the iteration magnitude. The spatial size of input data was set as 10×10 empirically. According to the size of samples, the proposed network was repetitively trained 300 epochs and 500 epochs on the eight subdatasets and other datasets to avoid overfitting and ensure convergence. All data were shuffled after each epoch to provide more abundant feature combinations. Besides, all independent experiments were conducted ten times to mitigate the randomness.

Moreover, the experiments were carried out on Windows OS with NVIDIA RTX 2080 GPU, Intel Core i7-8700K@3.4 GHz, and 32 GB of RAM. The basic DL framework used in the experiments was Pytorch 1.6.0, with Cuda 10.2 and cudnn 7.0.

Some metrics are selected for the performance evaluation of the proposed network and comparison methods. First, the inversion curves and map of Leaves-1 and Leaves-2 are used to evaluate the performance of RWCR qualitatively, and the root mse (RMSE) is used on the Lopex dataset for quantitative evaluation. The RMSE is calculated as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2} \tag{18}$$

where each symbol in (18) is exactly the same as that in (17). Second, OA, average accuracy (AA), average precision (AP), and kappa coefficient (κ) [70] based on the confusion matrix are used for the evaluation of classification performance. The n-class confusion matrix is shown in Fig. 14. There are n classes $C = \{C_1, C_2, \ldots, C_n\}$ in the confusion matrix, and the accuracy of the ith class is

$$P_{\mathrm{accuracy}}(C_i) = p(C_{ii}|C_i) = \frac{v_{ii}}{\sum_{j=1}^{n} v_{ij}}. \tag{19}$$

Similarly, the precision of the ith class is

$$P_{\mathrm{precision}}(C_i) = p(C_{ii}|C_i) = \frac{v_{ii}}{\sum_{j=1}^{n} v_{ji}}. \tag{20}$$

Here, P_{accuracy} means the probability of the ith classes classified correctly, which represents the classification capability of the model for the ith classes. $P_{\mathrm{precision}}$ means the ratio of the ith classes in the results labeled as C_i, which indicates the

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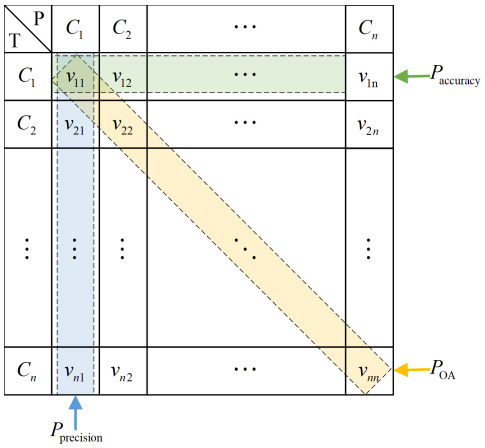

Fig. 14. Confusion matrix of n classes.

classification robustness of the model between C_i and others in $C = \{C_1, C_2, \ldots, C_{i-1}, C_{i+1}, \ldots, C_n\}$. Thus, P_{accuracy} represents the interclass accuracy, and $P_{\text{precision}}$ represents the intraclass precision of the model so that AA and AP indicate the interclass and intraclass classification capabilities of the model. Therefore, in the evaluation metrics of refined classification, OA is adopted to assess the overall classification capability of the model, AA and AP are adopted for evaluation of average classification performance of interclass and intraclass, and kappa coefficient is adopted for the agreement of classification results and truth labels.

IV. Experimental Results and Discussion

According to the experimental design of Section III, the experimental results and discussion are divided into the following four parts. The first two parts evaluate the performance of the subnet and key modules of the main net. The last two parts compare the proposed network with typical DL algorithms and some state-of-the-art HSI classification methods to verify the superiority.

A. Subnet Evaluation

The performance evaluation of the subnet is based on Leaves-1 and Leaves-2 datasets for the qualitative analysis and the Lopex dataset for the quantitative analysis.

In the qualitative evaluation of Leaves-1 and Leaves-2, all water contents retrieved are marked with relative labels from high to low levels. The highest level of water content is marked as Label:1, the second level of water content is marked as Label:2, and so on. The results of RWCR on five average spectral data of Leaves-1 are shown in Fig. 15, and the results on Leaves-2 are shown in Fig. 16. It is obvious that the results of RWCR in Fig. 15 are consistent with the known RWC of Leaves-1 described in Section III. Label:1 and Label:5 correspond to the leaves with the highest- and lowest-level RWC, namely, Fig. 8(a) and (d), respectively. The values of labels decrease with the increase in RWC. In Fig. 16, the inversion map of RWCR is consistent with the RWC observation of Fig. 9. It is obvious that the green leaves have

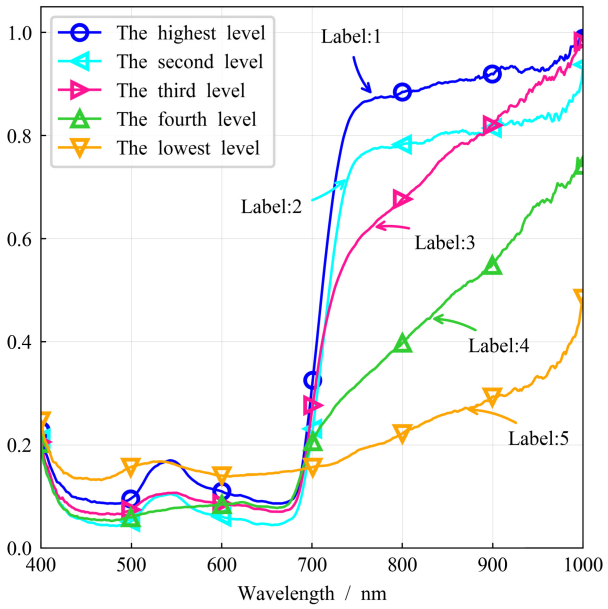

Fig. 15. RWCR results of average spectral data on Leaves-1.

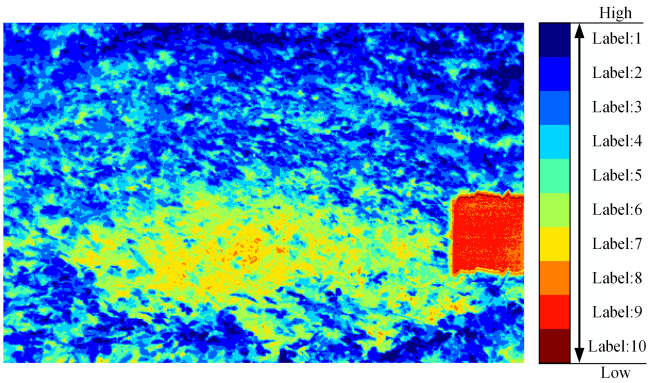

Fig. 16. Inversion map of RWCR results on Leaves-2.

the highest RWC, dry-yellow leaves haves a lower RWC, and the whiteboard for spectrum calibration has the lowest RWC.

In the quantitative evaluation on Lopex, 16 RWCR methods based on spectral indices are compared with the subnet, and RMSE is taken as the quantitative criterion. The RWCR methods contain green index (GI), water index (WI), SWIR water stress index (SIWSI), normalized difference infrared index (NDII), moisture stress index (MSI), normalized difference WI (NDWI), absorption depth ratio (ADR), normalized differential absorption depth index (NDADI), absorption depth (AD), and absorption symmetry (AS). Table VII shows the spectral indices, the corresponding formulations, and the RMSE results of RWCR. The RMSE value of the subnet in the proposed network is 0.80×10^{-6}, which is much smaller than the values of other typical RWCR methods.

Therefore, the qualitative and quantitative analyses show that the subnet has the capacity to extract accurate RWC features, and it is conducive to the expression of subtle RWC features in the proposed network.

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TABLE VII
COMPARISON RESULTS OF RWCR ON LOPEX

No.	Indices	Formulation	RMSE
1	GI_1	$f = 0.001446 \times GI_1 + 0.007678$	0.69×10^{-2}
2	GI_2	$f = -0.004303 \times GI_2 + 0.01536$	0.62×10^{-2}
3	WI_1	$f = 0.3557 \times WI_1 - 0.3511$	0.41×10^{-2}
4	WI_2	$f = 0.3827 \times WI_2 - 0.3764$	0.33×10^{-2}
5	$SIWSI_1$	$f = 0.08209 \times SIWSI_1 - 0.002692$	0.37×10^{-2}
6	$SIWSI_2$	$f = 0.08107 \times SIWSI_2 + 0.0007571$	0.30×10^{-2}
7	NDII	$f = 0.07847 \times NDII - 0.001594$	0.39×10^{-2}
8	MSI	$f = 0.03961 \times MSI + 0.03841$	0.47×10^{-2}
9	NDWI	$f = 0.2015 \times NDWI + 0.004971$	0.49×10^{-2}
10	ADR_1	$f = 0.002328 \times ADR_1 - 2.245$	0.38×10^{-2}
11	ADR_2	$f = 0.001992 \times ADR_2^{1.71}$	0.28×10^{-2}
12	$NDADI_1$	$f = 0.1049 \times NDADI_1^{2.008}$	0.38×10^{-2}
13	$NDADI_2$	$f = 0.001174 \times e^{4.882 \times NDADI_2}$	0.28×10^{-2}
14	AD	$f = 0.0008696 \times e^{4.56D}$	0.19×10^{-2}
15	AS	$f = 6.5\times10^{-3} \times S^{-2.197}$	0.23×10^{-2}
16	Proposed	Subnet	0.80×10^{-6}

TABLE VIII
CLASSIFICATION RESULTS OF THE PROPOSED NETWORK

Criteria	Class	OA (%)	AA (%)	AP (%)	κ
C-1	Corn	92.54	92.37	92.97	0.8874
	Lettuce	**96.79**	**96.83**	**96.76**	**0.9572**
C-2	**Grass**	**93.37**	**93.48**	**93.71**	**0.8999**
	BW	90.43	90.07	92.03	0.8070
C-3	**HOW**	**95.36**	**95.25**	**96.47**	**0.9302**
	HSW	92.71	92.29	92.09	0.8895
C-4	**MTB**	**88.58**	**88.59**	**90.46**	**0.8288**
	MTS	83.58	83.60	83.35	0.7537

TABLE IX
CLASSIFICATION RESULTS OF THE PROPOSED NETWORK WITHOUT ATTENTION

Criteria	Class	OA (%)	AA (%)	AP (%)	κ
C-1	Corn	91.62	91.53	92.07	0.8737
	Lettuce	**94.83**	**94.94**	**94.78**	**0.9311**
C-2	Grass	89.61	89.54	89.73	0.8429
	BW	90.07	89.90	90.44	0.8005
C-3	HOW	92.48	92.37	92.56	0.8871
	HSW	88.75	88.11	87.84	0.8297
C-4	**MTB**	**79.51**	**79.45**	**79.65**	**0.6926**
	MTS	79.29	79.49	80.19	0.6908

TABLE X
CLASSIFICATION RESULTS OF THE PROPOSED NETWORK WITHOUT RWCR

Criteria	Class	OA (%)	AA (%)	AP (%)	κ
C-1	Corn	89.97	89.88	90.19	0.8490
	Lettuce	**94.00**	**94.16**	**94.61**	**0.9200**
C-2	**Grass**	**80.77**	**80.01**	**80.63**	**0.7091**
	BW	80.64	80.76	80.95	0.6133
C-3	HOW	93.84	93.78	94.81	0.9075
	HSW	85.46	84.87	85.37	0.7798
C-4	MTB	82.33	82.32	84.48	0.7349
	MTS	77.68	77.45	78.09	0.6650

B. Ablation Study

The ablation study is carried out based on the trained subnet and contains three parts: the effectiveness test of key modules, the influence test of different spatial sizes, and the influence test of training sample size. The three parts of the experiments are conducted on the eight subdatasets described in Section III.

1) Key Modules: In this part, we have evaluated the improvements of the multiple-attention mechanism and the subnet based on FE units of MRFs. The results of this part are shown in Tables VIII–X, and the criteria of subdatasets are marked as C1–C4. C1 represents the same plants of pure species, C2 represents the same plants of mixed species, C3 represents the mixed plants, and C4 represents plants and nonplants. The datasets based on C1–C4 are prepared for the classification performance evaluation of interclass and intraclass.

Table VIII shows the classification results of the proposed network. For comparison, Tables IX and X show the classification results of the proposed network without the multiple-attention mechanism and the subnet. It is obvious that Table VIII achieves the best results of OA, AA, AP, and κ on each dataset. In Table IX, the best classification results are achieved on the Lettuce dataset with OA, AA, AP, and κ being 94.83%, 94.94%, 94.78%, and 0.9311, respectively. The corresponding results of the proposed network in Table VIII are 96.79%, 96.83%, 96.76%, and 0.9572, respectively. Therefore, the improvements of OA, AA, AP, and κ achieved on Lettuce by the multiple-attention mechanism are +1.96%, +1.89%, +1.98%, and +0.0261. The maximum improvements between Tables VIII and IX are achieved on the MTB dataset, which are +9.07%, +9.14%, +10.81%, and +0.1362, respectively. Similarly, in Table X, the best classification results are achieved on the Lettuce dataset with OA, AA, AP, and κ being 94.00%, 94.16%, 94.61%, and 0.9200, respectively. The corresponding results of the proposed network in Table VIII are 96.79%, 96.83%, 96.76%, and 0.9572, respectively. Therefore, the improvements of OA, AA, AP, and κ achieved on Lettuce by the subnet are +2.79%, +2.67%, +2.15%, and +0.0372. The maximum improvements between Tables VIII and X are achieved on the Grass dataset, which are +12.60%, +13.47%, +13.08%, and +0.1908, respectively.

To further illustrate the effectiveness of the subnet for preserving the subtle features, the feature maps of the subnet on IP, SV, and PU datasets are shown in Fig. 17. We had randomly selected a spectral curve from each data. The corresponding feature values of the 3-D RB and the subnet are normalized and shown with the spectral curves. The different feature values of 3-D RB and the subnet indicate their corresponding contributions for classification. As shown in Fig. 17, it is obvious that the contributions of the subnet are greater than that of the 3-D RB in the bands with subtle spectral variation (485–528 nm on IP, about 1000 nm on SV, and 482–500 and 517–526 nm on PU). Moreover, it is also observed the feature values of the subnet make more contributions in the bands of water absorption (1430–1450 and 1930–1970 nm on IP and SV, respectively). Thus, the subnet has the ability to effectively preserve the subtle and water-related features.

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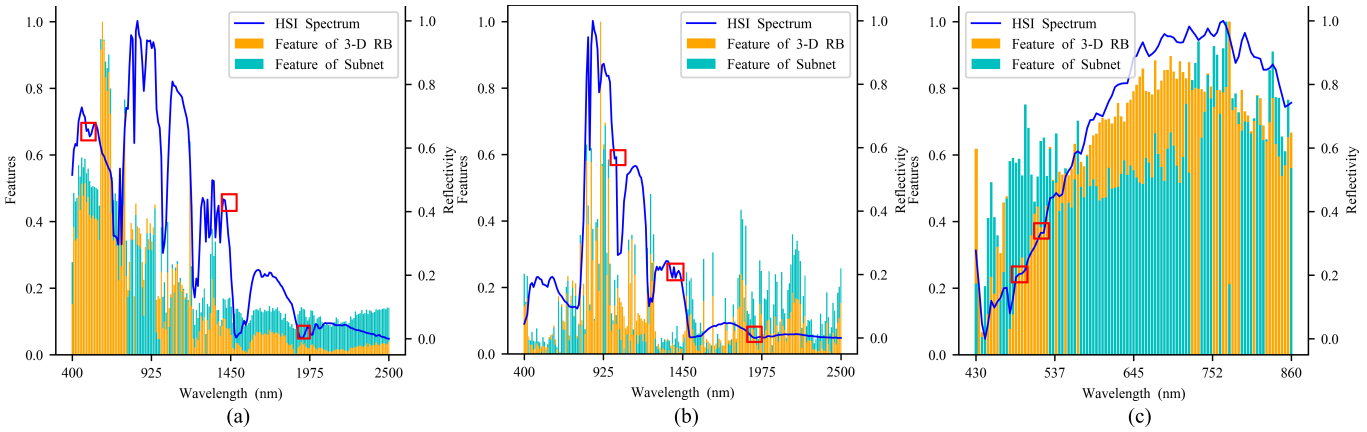

Fig. 17. Spectra and corresponding feature curves on (a) IP, (b) SV, and (c) PU datasets.

TABLE XI
CLASSIFICATION RESULTS OF DIFFERENT SPATIAL SIZES

Spatial Size	OA (%)	AA (%)	AP (%)	κ
8×8	89.67	90.00	89.68	0.8449
9×9	90.89	90.98	91.02	0.8630
10×10	**92.54**	**92.37**	**92.97**	**0.8874**
11×11	91.01	91.12	90.98	0.8648
12×12	90.60	90.83	90.58	0.8588

TABLE XII
CLASSIFICATION RESULTS OF DIFFERENT SIZES OF TRAINING SAMPLES

Training Size	OA (%)	AA (%)	AP (%)	κ
30	82.03	82.10	82.64	0.7297
40	86.77	86.60	88.12	0.8001
50	92.54	92.37	92.97	0.8874
60	93.49	93.56	93.57	0.9021
70	**95.19**	**95.35**	**95.21**	**0.9277**

Therefore, this part of the ablation experiments demonstrates the effectiveness of the RWCR and the multiple attentions based on the MRFs of the proposed network. Moreover, the classification accuracy of the proposed network on the same-species dataset, the same plants of mixed species, mixed-plants datasets, or mixed-ground-objects dataset is also improved. Thus, the accuracy improvements of interclass and intraclass on various datasets are demonstrated by AA and AP, and the overall classification performances are demonstrated by OA and κ. Therefore, the proposed network extracts representative and effective interclass features and enhances the expression of the subtle diagnostic intraclass features on small-sized data, which improves the accuracy of interclass/intraclass and the performance of refined classification.

2) Spatial Sizes: In this part, we have evaluated the effect of different spatial sizes on the Corn dataset. The spatial size of input data determines the amount of spatial information input to the model. Empirically, five different spatial sizes were selected for testing, containing 8×8, 9×9, 10×10, 11×11, and 12×12. The classification results are shown in Table XI. It is observed that the classification accuracy varies with the spatial size, and the spatial size 10×10 achieves the best performance, which is consistent with the default value of experiments in this article.

3) Training Sizes: Similarly, we also have evaluated the effect of different training sizes of input on the Corn dataset. The training size is an important factor affecting the accuracy of FE and classification result in DL methods. We empirically select five different sizes of input data for training, which are 30, 40, 50, 60, and 70. The classification results are shown in Table XII. It is obvious that the classification accuracy increases with the training size, and the 70 labeled samples for training achieve the best classification performance.

Therefore, the classification performance of the proposed network is affected by the spatial size and the training number of the input samples. The empirically selected spatial size 10×10 obtains the highest accuracy on OA, AA, AP, and κ. As with other DL methods, the larger the size of training samples, the higher the accuracy of classification. Moreover, the ablation study results indicate that the proposed network has the capacity to extract RWC features, increase the classification accuracy of interclass/intraclass, and improve the overall performance of refined classification.

C. Comparison With the Typical DL Algorithms

In order to evaluate the superiority of the proposed network, some typical DL algorithms described in the experimental design of Section III, containing AlexNet, DenseNet, GoogleNet, Inception V3, and ResNet, are compared with the proposed network. One dataset is selected according to each criterion so that four datasets (Corn, Grass, HOW, and MTB) are taken for training, validation, and testing. The classification results are shown in Tables XIII–XVI.

The results in Tables XIII–XVI show that the proposed network achieves the best classification performance than others. In Table XIII, DenseNet achieves the second-best performance, and Inception V3 archives the lowest performance on the Corn dataset. That is, the minimum accuracy improvements of the proposed network over other DL methods are +4.88%, +1.72%, +10.01%, and +0.1093 for OA, AA, AP, and κ, respectively, and the maximum accuracy improvements are +7.32%, +4.06%, +15.08%, and

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TABLE XIII
CLASSIFICATION RESULTS OF DL METHODS ON THE CORN DATASET

	OA (%)	AA (%)	AP (%)	κ
AlexNet	85.56	87.10	77.24	0.7443
DenseNet	87.66	90.65	82.96	0.7781
GoogleNet	86.68	89.26	81.31	0.7601
Inception V3	85.22	88.31	77.89	0.7389
ResNet	86.49	88.27	81.65	0.7551
Proposed Net	**92.54**	**92.37**	**92.97**	**0.8874**

TABLE XIV
CLASSIFICATION RESULTS OF DL METHODS ON THE GRASS DATASET

	OA (%)	AA (%)	AP (%)	κ
AlexNet	77.53	84.00	62.71	0.5775
DenseNet	79.58	82.96	63.49	0.6104
GoogleNet	76.89	84.04	65.08	0.5534
Inception V3	69.68	78.37	56.20	0.4456
ResNet	80.42	84.75	74.45	0.6058
Proposed Net	**93.37**	**93.48**	**93.71**	**0.8999**

TABLE XV
CLASSIFICATION RESULTS OF DL METHODS ON THE HOW DATASET

	OA (%)	AA (%)	AP (%)	κ
AlexNet	88.79	91.85	67.60	0.7548
DenseNet	91.52	94.57	67.60	0.8114
GoogleNet	90.30	93.32	66.68	0.7839
Inception V3	90.99	86.46	66.39	0.7952
ResNet	92.16	94.82	68.01	0.8230
Proposed Net	**95.36**	**95.25**	**96.47**	**0.9302**

TABLE XVI
CLASSIFICATION RESULTS OF DL METHODS ON THE MTB DATASET

	OA (%)	AA (%)	AP (%)	κ
AlexNet	78.92	83.87	71.58	0.5944
DenseNet	82.79	81.86	76.62	0.6515
GoogleNet	79.66	81.55	73.13	0.6082
Inception V3	77.14	78.24	71.92	0.5630
ResNet	82.41	80.29	72.06	0.6422
Proposed Net	**88.58**	**88.59**	**90.46**	**0.8288**

0.1485, respectively. Different from Table XIII, Table XIV shows that ResNet achieves the second-best performance, and Inception V3 archives the lowest performance on the Grass dataset. That is, the minimum accuracy improvements of the proposed network over other DL methods are +12.95%, +8.73%, +19.26%, and +0.2941 for OA, AA, AP, and κ, respectively, and the maximum accuracy improvements are +23.69%, +15.11%, +37.51%, and 0.4543, respectively. Similar to Table XIV, Table XV shows that ResNet achieves the second-best performance and AlexNet archives the lowest performance on the HOW dataset. That is, the minimum accuracy improvements of the proposed network over other DL methods are +3.20%, +0.43%, +28.46%, and +0.1072 for OA, AA, AP, and κ, respectively, and the maximum accuracy improvements are +6.57%, +3.40%, +28.87%, and +0.1754, respectively. Table XVI shows that DenseNet achieves the second-best performance on the MTB dataset, which is the same as that in Table XIII. Inception V3 archives the lowest performance. That is, the minimum accuracy improvements of the proposed network over other DL methods are +5.79%, +6.73%, +13.84%, and +0.1773 for OA, AA, AP, and κ, respectively, and the maximum accuracy improvements are +11.44%, +10.35%, +18.54%, and +0.2658, respectively.

The improvements exhibited in Tables XIII–XVI suggest that the proposed network can extract more accurate spectral–spatial features than other typical DL algorithms. The overall refined classification performance is improved to different extents on each dataset. Especially, the improvements of AP are much more considerable than OA, AA, and κ, which indicates that the proposed network not only extracts representative and effective features to improve interclass accuracy but also enhances the expression of the subtle RWC features to improve intraclass accuracy in classification.

D. Comparison With HSI Classification Methods

In order to further evaluate the superiority of the proposed network in overall classification performance, this part contains four groups of experiments. As described in the experimental settings of Section III, the first group is the comparison with some traditional classification methods, containing SVM, RF, MLP, 2-D CNN, and 3-D CNN. All of the methods were trained base on 15% labeled samples according to the corresponding literature, and the classification results are shown in Tables XVII–XIX. The second group is the comparison with some state-of-the-art approaches, containing SMBN, DFFN, DHCNet, SSRN, SSN, and SSAN. Moreover, the proposed networks without the RWCR module [proposed-non-RWCR (PNR)] and the attention module [proposed-nonattention (PNA)] are also compared with the proposed network. All of the approaches were trained based on 10% IP, 2% SV, and 2% PU according to the corresponding literature. For a fair comparison, all of the experimental settings are keeping the same. The classification results are shown in Tables XX–XXII. The third group is the comparison with some models for small samples classification based on five labeled training samples per class of each dataset, which contains SVM, 3-D CNN, SSRN, RN-FSC, and DCFSL. The classification results are shown in Table XXIII. To further validate the superiority of the proposed network to other methods, the fourth group selects the above SVM, 2-D CNN, 3-D CNN, SSRN, and SSAN for the comparison experiment on the Houston dataset. All of the methods were trained based on 10%, 20%, and 30% of the training dataset. Other experimental settings are keeping the same, and the classification results are shown in Table XXIV.

In the first group of experiments, Table XVII shows the classification results on the IP dataset. The proposed network obtains the highest accuracy in each class and archives 99.55% of OA, 99.63% of AA, and 0.9953 of κ, respectively. The accuracy improvements of the proposed network over the second-best method (3-D CNN) are +2.65%, +6.29%, and +0.0636, respectively. Similarly, the proposed network achieves the best performance on SV and PU datasets. Table XVIII shows the classification results on the SV dataset. The proposed network achieves the highest accuracy in each class and obtains 99.92% of OA, 99.93% of AA, and 0.9991 of κ.

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TABLE XVII
CLASSIFICATION RESULTS ON THE IP DATASET (15% LABELED SAMPLES FOR TRAINING)

No.	Ground-truth type	SVM	RF	MLP	2-D CNN	3-D CNN	Proposed
1	Alfalfa	65.64	31.28	61.79	64.62	87.95	**100.00**
2	Corn-notill	83.45	66.41	83.80	80.85	96.91	**99.68**
3	Corn-mintill	74.70	55.21	82.68	78.34	97.22	**99.88**
4	Corn	79.25	39.95	62.84	80.85	95.57	**100.00**
5	Grass-pasture	88.90	85.73	90.27	68.20	94.78	**100.00**
6	Grass-trees	93.50	94.79	94.27	87.29	98.42	**99.86**
7	Grass-pasture-mowed	66.09	31.30	70.43	63.04	86.09	**100.00**
8	Hay-windrowed	95.42	98.15	94.31	96.67	99.16	**100.00**
9	Oats	54.12	12.94	68.24	72.94	67.65	**100.00**
10	Soybean-notill	79.49	70.01	82.09	68.90	96.94	**99.79**
11	Soybean-mintill	87.41	90.56	83.37	76.09	97.38	**98.80**
12	Soybean-clean	93.79	54.72	75.71	96.69	92.36	**99.83**
13	Wheat	72.93	92.53	97.82	96.67	98.51	**100.00**
14	Woods	95.40	96.02	94.99	95.75	99.10	**99.45**
15	Buildings-Grass-Trees-Drives	75.09	50.85	64.85	95.55	94.57	**100.00**
16	Stone-Steel-Towers	91.90	86.46	87.97	91.01	90.89	**100.00**
	OA (%)	86.19	78.17	84.86	82.56	96.90	**99.55**
	AA (%)	81.07	66.06	80.96	82.09	93.34	**99.63**
	κ	0.8075	0.6583	0.7916	0.8124	0.9317	**0.9953**

TABLE XVIII
CLASSIFICATION RESULTS ON THE SV DATASET (15% LABELED SAMPLES FOR TRAINING)

No.	Ground-truth type	SVM	RF	MLP	2-D CNN	3-D CNN	Proposed
1	Brocoli_green_weeds_1	98.97	99.48	98.90	87.85	99.72	**100.00**
2	Brocoli_green_weeds_2	99.08	99.58	99.55	99.48	99.77	**100.00**
3	Fallow	97.12	98.59	97.43	80.98	98.74	**100.00**
4	Fallow_rough_plow	98.13	98.76	97.96	95.00	98.78	**99.78**
5	Fallow_smooth	98.36	98.24	98.84	63.25	99.29	**99.81**
6	Stubble	99.33	99.48	99.74	99.33	99.78	**99.95**
7	Celery	99.42	99.14	99.13	97.89	99.59	**99.97**
8	Grapes_untrained	91.07	84.34	85.34	91.75	90.59	**99.95**
9	Soil_vinyard_develop	99.75	98.91	99.76	97.86	99.71	**100.00**
10	Corn_senesced_green_weeds	98.98	93.34	96.97	96.86	98.34	**99.79**
11	Lettuce_romaine_4wk	99.54	95.62	96.64	93.64	99.18	**100.00**
12	Lettuce_romaine_5wk	99.08	97.84	99.32	92.16	99.32	**100.00**
13	Lettuce_romaine_6wk	98.95	97.51	99.81	97.66	98.44	**99.95**
14	Lettuce_romaine_7wk	97.60	94.20	94.57	94.38	99.30	**100.00**
15	Vinyard_untrained	72.65	63.55	79.92	86.75	96.06	**99.77**
16	Vinyard_vertical_trellis	99.63	97.89	98.35	93.15	98.93	**100.00**
	OA (%)	93.83	90.61	93.43	91.97	97.08	**99.92**
	AA (%)	96.73	94.78	96.39	91.75	98.47	**99.93**
	κ	0.9321	0.8979	0.9286	0.9019	0.9673	**0.9991**

TABLE XIX
CLASSIFICATION RESULTS ON THE PU DATASET (15% LABELED SAMPLES FOR TRAINING)

No.	Ground-truth type	SVM	RF	MLP	2-D CNN	3-D CNN	Proposed
1	Asphalt	94.61	93.52	93.81	92.82	98.98	**99.82**
2	Meadows	97.87	98.29	97.98	96.85	99.59	**99.97**
3	Gravel	83.11	29.83	79.05	91.00	96.35	**100.00**
4	Trees	95.43	91.68	92.98	94.42	98.65	**99.53**
5	Painted metal sheets	99.12	98.88	89.76	89.55	99.32	**99.67**
6	Bare Soil	91.86	74.54	91.94	89.07	98.65	**100.00**
7	Bitumen	88.74	81.01	87.32	79.99	90.79	**100.00**
8	Self-Blocking Bricks	89.67	90.70	89.37	96.27	98.34	**99.88**
9	Shadows	**99.78**	99.75	96.62	96.07	99.58	99.84
	OA (%)	94.85	89.79	93.98	94.03	98.77	**99.90**
	AA (%)	93.35	84.24	90.98	91.78	97.81	**99.87**
	κ	0.9224	0.8216	0.9037	0.9067	0.9752	**0.9987**

The accuracy improvements of the proposed network over the second-best method (3-D CNN) are +2.84%, +1.46%, and +0.0318, respectively. Table XIX shows the classification results on the PU dataset, and the proposed network achieves the highest accuracy on most classes and obtains 99.90% of OA, 99.87% of AA, and 0.9987 of κ. The accuracy improvements of the proposed network over the second-best method (3-D CNN) are +1.13%, +2.06%, and +0.0235,

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TABLE XX
CLASSIFICATION RESULTS ON THE IP DATASET (10% LABELED SAMPLES FOR TRAINING)

No.	Ground-truth type	SMBN	DFFN	DHCNet	SSRN	SSN	SSAN	PNR	PNA	Proposed
1	Alfalfa	86.67	100.00	88.61	85.83	83.06	83.61	95.83	98.06	**100.00**
2	Corn-notill	83.93	92.15	92.05	**92.99**	91.56	91.70	90.11	91.16	91.81
3	Corn-mintill	94.22	95.11	89.82	98.01	94.79	94.74	92.24	92.70	**98.85**
4	Corn	85.87	96.51	84.18	64.02	75.77	90.79	96.61	98.57	**100.00**
5	Grass-pasture	99.40	98.73	97.87	98.00	96.05	99.12	95.90	98.05	**99.88**
6	Grass-trees	99.38	**99.98**	99.76	99.50	99.35	99.23	96.34	97.41	99.23
7	Grass-pasture-mowed	97.73	71.82	84.09	64.55	92.73	97.73	96.36	98.18	**100.00**
8	Hay-windrowed	97.36	99.87	98.51	**100.00**	99.11	98.93	98.22	99.27	**100.00**
9	Oats	**100.00**	90.63	88.75	46.88	95.63	**100.00**	98.75	99.38	**100.00**
10	Soybean-notill	86.40	95.67	92.94	92.46	91.30	95.01	93.75	94.91	**97.80**
11	Soybean-mintill	97.85	96.39	96.71	98.71	**99.11**	98.25	95.25	96.78	97.71
12	Soybean-clean	89.49	88.48	87.78	92.01	92.54	95.20	97.08	98.35	**99.15**
13	Wheat	95.77	98.47	99.33	**100.00**	97.30	**100.00**	98.04	99.26	**100.00**
14	Woods	98.19	98.74	97.87	**98.78**	98.61	98.62	95.12	97.20	97.73
15	Buildings-Grass-Trees-Drives	89.03	97.27	89.45	86.27	84.97	86.27	96.27	99.09	**100.00**
16	Stone-Steel-Towers	96.30	99.04	93.29	**100.00**	95.48	98.36	97.53	99.79	**100.00**
	OA (%)	93.54	96.03	94.50	95.49	95.16	96.11	94.64	96.07	**97.62**
	AA (%)	93.60	94.93	92.56	88.63	92.96	95.47	95.84	97.39	**98.89**
	κ	0.9219	0.9471	0.9183	0.8728	0.9176	0.9528	0.9438	0.9593	**0.9729**

TABLE XXI
CLASSIFICATION RESULTS ON THE SV DATASET (2% LABELED SAMPLES FOR TRAINING)

No.	Ground-truth type	SMBN	DFFN	DHCNet	SSRN	SSN	SSAN	PNR	PNA	Proposed
1	Brocoli_green_weeds_1	82.24	99.38	99.56	**100.00**	99.17	99.35	99.11	99.61	99.95
2	Brocoli_green_weeds_2	99.77	99.78	99.82	98.27	99.91	**99.95**	97.91	98.27	99.50
3	Fallow	90.40	96.62	98.14	**99.55**	98.54	99.06	99.11	99.26	99.53
4	Fallow_rough_plow	98.90	99.16	**99.98**	99.52	98.93	98.88	96.63	97.03	97.78
5	Fallow_smooth	98.27	99.40	98.35	99.43	98.53	**99.85**	98.83	98.94	99.38
6	Stubble	99.98	99.90	99.93	**100.00**	**100.00**	99.91	99.06	99.77	99.86
7	Celery	99.29	99.60	99.67	**100.00**	99.60	99.96	98.63	99.08	99.34
8	Grapes_untrained	89.26	**94.30**	89.41	90.71	92.01	92.91	92.86	92.98	93.92
9	Soil_vinyard_develop	99.41	98.96	**100.00**	**100.00**	99.61	99.95	97.51	98.37	98.94
10	Corn_senesced_green_weeds	96.82	95.98	97.90	98.05	98.01	**98.91**	97.87	98.05	98.79
11	Lettuce_romaine_4wk	98.79	98.37	98.90	**99.80**	99.46	99.21	97.53	97.94	98.07
12	Lettuce_romaine_5wk	97.26	**100.00**	99.60	**100.00**	99.10	**100.00**	97.92	98.87	99.36
13	Lettuce_romaine_6wk	99.40	99.53	98.42	**100.00**	**100.00**	**100.00**	95.96	96.51	96.56
14	Lettuce_romaine_7wk	98.50	99.40	**100.00**	99.20	98.50	99.78	96.20	96.94	97.06
15	Vinyard_untrained	92.20	88.46	89.71	93.60	90.67	91.03	93.60	95.58	**95.77**
16	Vinyard_vertical_trellis	98.37	99.21	98.40	99.32	98.71	98.40	98.68	99.37	**99.43**
	OA (%)	95.04	96.57	95.96	96.86	96.58	97.06	96.41	97.04	**97.58**
	AA (%)	96.18	98.00	97.99	**98.59**	98.17	98.57	97.34	97.91	98.33
	κ	0.9486	0.9541	0.9482	0.9538	0.9557	0.9689	0.9623	0.9698	**0.9730**

TABLE XXII
CLASSIFICATION RESULTS ON THE PU DATASET (2% LABELED SAMPLES FOR TRAINING)

No.	Ground-truth type	SMBN	DFFN	DHCNet	SSRN	SSN	SSAN	PNR	PNA	Proposed
1	Asphalt	94.90	94.65	96.18	97.90	98.24	**98.74**	96.18	96.25	96.27
2	Meadows	**99.64**	99.11	98.93	98.65	98.98	99.33	99.24	99.34	99.55
3	Gravel	80.05	89.46	87.90	85.87	84.98	97.10	98.01	98.17	**98.21**
4	Trees	96.17	98.13	**99.76**	99.12	99.05	97.32	94.97	95.09	95.12
5	Painted metal sheets	94.81	99.14	**99.98**	99.48	97.95	96.10	99.72	99.82	99.90
6	Bare Soil	97.77	95.40	94.83	97.03	**99.65**	97.78	98.11	98.25	98.52
7	Bitumen	88.02	92.61	96.18	96.79	97.73	97.02	98.33	98.56	**98.88**
8	Self-Blocking Bricks	84.91	93.51	94.43	**97.46**	94.12	94.00	96.63	97.01	97.17
9	Shadows	99.06	98.82	98.12	**100.00**	**100.00**	**100.00**	92.27	94.42	94.55
	OA (%)	95.68	96.75	97.08	97.65	97.79	98.19	97.87	98.05	**98.21**
	AA (%)	92.81	95.65	96.26	96.92	96.74	97.49	97.05	97.43	**97.57**
	κ	0.9216	0.9531	0.9577	0.9689	0.9582	0.9738	0.9659	0.9737	**0.9763**

respectively. Therefore, in the comparison with traditional classification methods, the experimental results demonstrated that the proposed network is superior to others in the overall classification performance.

In the second group of experiments, Table XX shows the classification results on the IP dataset, and the corresponding classification maps are shown in Fig. 18. The proposed network achieves the highest accuracy in most classes and obtains

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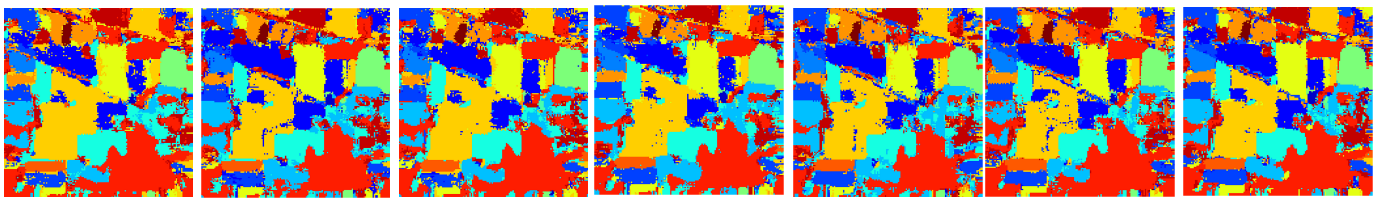
TABLE XXIII
CLASSIFICATION RESULTS ON DATASETS (FIVE LABELED SAMPLES PER CLASS)

Method	IP			SV			PU		
	OA (%)	AA (%)	κ	OA (%)	AA (%)	κ	OA (%)	AA (%)	κ
SVM	45.85	59.24	0.3968	80.71	87.58	0.7861	64.12	68.18	0.5559
3-D CNN	54.76	63.93	0.4872	84.20	89.56	0.8246	65.74	73.72	0.5737
SSRN	61.36	59.75	0.5691	86.39	93.24	0.8495	76.26	79.51	0.7056
RN-FSC	58.17	69.90	0.5252	84.11	88.83	0.8238	80.19	77.12	0.7373
DCFSL	66.81	77.89	0.6264	89.34	**94.04**	0.8817	83.65	83.77	0.7870
Proposed	**70.80**	**81.69**	**0.6690**	**91.74**	93.76	**0.9081**	**87.51**	**88.92**	**0.8384**

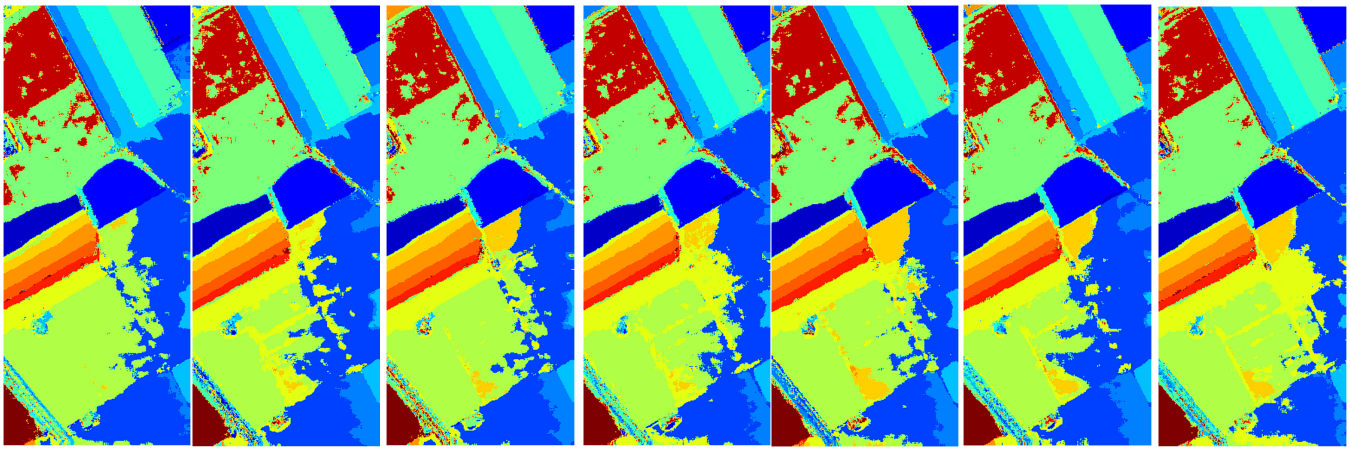
TABLE XXIV
CLASSIFICATION RESULTS ON THE HOUSTON DATASET

Method	10%			20%			30%		
	OA (%)	AA (%)	κ	OA (%)	AA (%)	κ	OA (%)	AA (%)	κ
SVM	63.15	61.02	0.6073	66.48	63.78	0.6125	74.78	73.29	0.6939
2-D CNN	63.61	61.14	0.6089	66.75	63.70	0.6163	69.41	67.57	0.6587
3-D CNN	75.69	74.02	0.7387	77.45	75.82	0.7507	79.65	77.18	0.7594
SSRN	78.96	78.57	0.7832	80.19	77.41	0.7715	82.22	80.03	0.7987
SSAN	79.10	78.02	0.7796	81.04	79.65	0.7886	83.78	81.72	0.8059
Proposed	**81.77**	**79.14**	**0.7903**	**82.67**	**80.27**	**0.8031**	**84.54**	**83.02**	**0.8268**



(a) (b) (c) (d) (e) (f) (g)

Fig. 18. Classification maps on the IP dataset. (a) SMBN (93.54%). (b) DFFN (96.03%). (c) DHCNet (94.50%). (d) SSRN (95.49%). (e) SSN (95.16%). (f) SSAN (96.11%). (g) Proposed (97.62%).



(a) (b) (c) (d) (e) (f) (g)

Fig. 19. Classification maps on the SV dataset. (a) SMBN (95.04%). (b) DFFN (96.57%). (c) DHCNet (95.96%). (d) SSRN (96.86%). (e) SSN (96.58%). (f) SSAN (97.06%). (g) Proposed (97.58%).

97.62% of OA, 98.89% of AA, and 0.9729 of κ. The accuracy improvements of the proposed network over the second-best method (SSAN) are +1.51%, +3.42%, and +0.0201, respectively. Table XXI shows the classification results on the SV dataset, and the corresponding classification maps are shown in Fig. 19. The proposed network has not achieved the highest accuracy in all classes, and SSRN achieves the best AA (98.59%), but the proposed method achieves the highest OA and κ, which are 97.58% and 0.9730. The improvements of OA and κ over the second-best method (SSAN) are +0.52% and +0.0041, respectively. Table XXII shows the classification results on the PU dataset, and the corresponding classification maps are shown in Fig. 20. Similar to Table XXI, the proposed network has not achieved the highest accuracy in all classes

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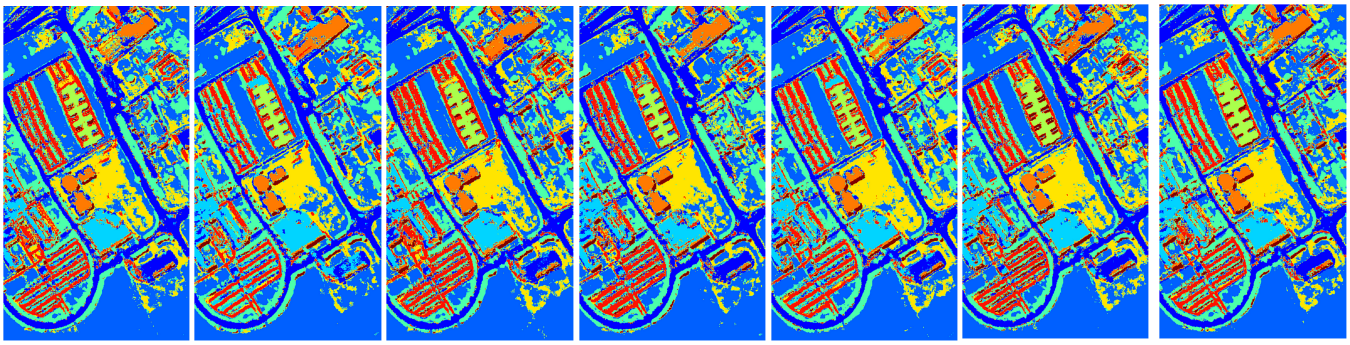

Fig. 20. Classification maps on the PU dataset. (a) SMBN (95.68%). (b) DFFN (96.75%). (c) DHCNet (97.08%). (d) SSRN (97.65%). (e) SSN (97.79%). (f) SSAN (98.19%). (g) Proposed (98.21%).

of PU, but it achieves the highest OA, AA, and κ, which are 98.21%, 97.57%, and 0.9763, respectively. The accuracy improvements of the proposed network over the second-best method (SSAN) are +0.02%, +0.08%, and +0.0025, respectively. Moreover, the proposed network shows significant improvements over the PNR and PNA models on IP, SV, and PU datasets. The improvements of OA over the PNR and PNA are +2.98% and +1.55% on IP, +1.17% and +0.54% on SV, and +0.34% and +0.16% on PU, respectively, which demonstrates the effectiveness of the RWCR and attention modules. Therefore, in the comparison with the state-of-the-art HSI classification methods, the superiority of the proposed network in overall classification performance is demonstrated on IP, SV, and PU datasets.

In the third group of experiments, Table XXIII shows the classification results on five labeled training samples of IP, SV, and PU datasets. The proposed network achieves the best performance on all datasets. The proposed network obtains 70.80%, 91.74%, and 87.51% of OA on IP, SV, and PU datasets, respectively. The improvements of the proposed network over the second-best method (DCFSL) are +3.99%, +2.40%, and +3.86%, respectively. Similarly, the proposed network achieves 81.69%, 93.76%, and 88.92% of AA and 0.6690, 0.9081, and 0.8384 of κ. The AA of the DCFSL on SV datasets achieves the best accuracy, but the AA improvements of the proposed network on IP and PU datasets over the DCFSL are +3.80% and +5.15%, respectively. Moreover, the κ improvements of the proposed network over the DCFSL are +0.0426, +0.0264, and +0.0514 on IP, SV, and PU datasets, respectively.

In the fourth group of experiments, Table XXIV shows the classification results on the Houston dataset. The proposed network achieves the highest accuracy on all sampling ratios of training sets. The proposed network obtains 81.77% of OA, 79.14% of AA, and 0.7903 of κ on 10% of the training set, which increases the improvements of accuracy over the second-best method (SSAN) are +2.67%, +1.12%, and +0.0107, respectively. Similarly, the proposed network obtains 82.67% of OA, 80.27% of AA, and 0.8031 of κ on 20% of the training set, which increases the improvements of accuracy over the second-best method (SSAN), which are +1.63%, +0.62%, and +0.0145, respectively. Moreover, the proposed network obtains 84.54% of OA, 83.02% of AA, and 0.8268 of κ on 30% of the training set, which increases the improvements of accuracy over the second-best method (SSAN) are +0.76%, +1.30%, and +0.0209, respectively. Therefore, it is also demonstrated on the Houston dataset that the classification performance of the proposed network is superior to that of other state-of-the-art models.

TABLE XXV
COMPUTATIONAL COMPLEXITY OF EACH DL-BASED MODEL

Methods	Parameters	FLOPs(G)	Training time (m)	Testing time (s)
2-D CNN	1937184	0.105	9.7	3.1
3-D CNN	2756164	0.1958	15.1	3.5
SMBN	321084	0.2208	18.3	4.7
DFFN	5224272	0.3714	34.6	7.5
DHCNet	629404	0.5252	42.7	9.6
SSRN	282232	1.3196	115.9	29.6
SSAN	314020	1.6309	121.3	35.4
Proposed	322712	2.6333	211.7	68.1

To analyze the computational complexity of the proposed network and other DL-based methods, the 2-D CNN, 3-D CNN, SMBN, DFFN, DHCNet, SSRN, and SSAN were selected for the comparison experiment with the proposed network. Only the core parts of FE in the above networks were chosen to calculate the corresponding number of network parameters, the number of the floating-point operations (FLOPs), and the average training and testing time (on GPU) in the third group of experiments. The calculation results are shown in Table XXV. It is observed that the parameters of each model are different, and the average time is positively correlated with the FLOPs. The time consumption of the proposed network is more than that of other methods, which means that the improvements of classification performance increase the computation complexity.

In summary, we have conducted four parts of experiments on 15 datasets to test the performance of the proposed network, and five metrics are adopted for quantitative evaluation. Specifically, RMSE is used for the effectiveness evaluation of the subnet, OA and κ are used for the performance evaluation of overall classification, and AA and AP are used for the performance evaluations of interclass and intraclass

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classifications, respectively. In the first part of the experiments, the performance of the subnet is tested on three leaves datasets, and the subnet exhibits the best performance than other methods. In the second ablation part, we have tested the effectiveness of the key modules and the influence of two hyperparameters (the spatial size and training size of input data) on eight subdatasets, and the maximum improvements of OA, AA, AP, and κ are +12.60%, +13.47%, +13.08%, and +0.1908, respectively. The significant improvements of key modules for classification are demonstrated, and the results of the refined classification show that the proposed network can not only extract representative and effective features for interclass classification but also enhance the expression of the subtle diagnostic features for intraclass classification. Besides, the comparison with some typical DL algorithms and some other HSI classification methods are carried out in the third and fourth parts based on four subdatasets and four public datasets (IP, SV, PU, and Houston). In the third part, the improvements of OA over the second-best DL methods are +4.88%, +12.95%, +3.20%, and +5.79% on four subdatasets, respectively, which demonstrates the superiority of the proposed network. In the fourth part, in comparison with traditional and state-of-the-art HSI classification methods on three public datasets, the maximum improvements of OA over the second-best methods in each group are +2.84%, +1.51%, +3.99%, and 2.67%, respectively, which demonstrates the competitive advantage of the proposed network. Although the classification performance has been improved, the proposed network still increases the computation complexity over other methods.

V. Conclusion

In this article, a water retrieval embedded attention network with MRFs is proposed for refined classification on small-sized datasets. There are three critical modules in the proposed network, namely, the subnet, multiple-attention mechanism, and MRFs. The subnet is trained on RWC datasets and responsible for enhancing the expression of subtle diagnostic features of RWC and improving the intraclass accuracy. The parameters of the subnet are not updated in the classification training of the main net. The multiple-attention mechanism and MRFs are responsible for extracting representative and effective features to improve the interclass accuracy. The multiple-attention mechanism contains band attention, channel attention, and scale attention, which weigh the corresponding features in classification. The MRFs are composed of three different scales of RFs, namely, RF-3, RF-5, and RF-7, which are the core part of FE. The experimental results verify the effectiveness of the critical modules and demonstrate the accuracy improvements of the proposed network in interclass/intraclass classification. Besides, the superiority of the proposed network to the DL methods and the state-of-the-art HSI classification methods is also demonstrated. That is, the proposed network has the capacity to extract representative features and subtle diagnostic RWC features from the small-sized datasets, which significantly improves the accuracy of interclass/intraclass in refined classification and achieves better performance than other classification methods.

Although the proposed network exhibits a significant improvement in refined classification, there are still some challenges that should be addressed. First, the accuracy of the subnet is an important factor affecting classification accuracy, which should be improved further. Second, the structure of the proposed network can be further simplified to reduce parameters and accelerate the training process.

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